Eidgenössische Technische Hochschule Zürich

Analysis I: One Variable

Exam Preperation Course Notes

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This script was created as part of a PVK course on Analysis I: One Variable in the fall semester 2024 at ETH Zurich and was later revised for the fall of 2025. It is based on the lecture by Prof. Figalli in HS2023 and aims to present the core of the Analysis I lecture at ETH in a concise and clear manner. Errors in formulation, presentation, and typos are possible, and improvements and corrections are warmly welcome. It is important to emphasize that this script is by no means complete, but is intended to serve as a guide for the PVK and exam preparation. Only theorems whose proofs are relevant for the exam are proved, and only the most important definitions and theorems are mentioned.

Good luck with your exam preparation!

Zürich, January 11, 2026

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Chapter 1

Basics, Sets and Sequences

1.1 Mathematical Induction

Mathematical induction is a proof principle used to show that a statement holds for all natural numbers. For example (see Example 1.1.1), one can show that

$$\sum_{k=1}^{n} k = 1 + 2 + \cdots + n = \frac{n(n+1)}{2} \qquad \text{for all } n \in \mathbb{N}.$$

In an induction proof, such a statement is proved in two steps.

- Base case: The statement is true for $n = 0$.
- Inductive step: If the statement is true for $n = k$, then it is also true for $k + 1$.

From these two statements, the principle of induction implies that the statement is true for all $n \in \mathbb{N}$.

We know the statement holds for $n = 0$ (base case). Because we have proved the inductive step, we now know the statement is also true for $n = 1$. This can be continued indefinitely, and thus the statement follows for all natural numbers. For the sake of completeness, it should also be noted that the base case does not always have to be $n = 0$. Any number from which one wants to show the statement can be chosen. If the statement should only hold for all $n \geq 2$, then $n = 2$ can be chosen as the base case.

Example 1.1.1 (Sum of Natural Numbers)
We prove by induction:

$$\sum_{k=1}^{n} k = 1 + 2 + \cdots + n = \frac{n(n+1)}{2} \qquad \text{for all } n \in \mathbb{N}.$$

Proof. We choose $n = 0$ as our *base case*. For $n = 0$ we have

$$0 = \frac{0 \cdot (0+1)}{2} = 0.$$

Now we show the inductive step. Let the statement be true for $n = k$ (*induction hypothesis*). That is, explicitly,

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}$$

Now consider the sum $1 + 2 + \cdots + k + (k+1)$. Using the *induction hypothesis*, we can write the sum as

$$\underbrace{1 + 2 + \cdots + k}_{=\frac{k(k+1)}{2}} + (k+1) = \frac{k(k+1)}{2} + (k+1).$$

We can now further simplify this expression

$$\frac{k(k+1)}{2} + (k+1) = \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2}.$$

This corresponds exactly to the statement to be proved for $n = k+1$. From the assumption that the statement holds for $n = k$, it follows that the statement also holds for $k+1$. Thus, the inductive step is proved. From the *base case* and the *inductive step*, it follows by mathematical induction that the statement holds for all $n \in \mathbb{N}$. □

1.2 Sequences

Definition 1.2.1 (Sequence)
A **sequence** is a function $x : \mathbb{N} \to \mathbb{R}$. The function value $x(n)$ is called the **n-th term** of the sequence and is denoted by x_n. Sequences are denoted as $(x_n)_{n\in\mathbb{N}}$ or often simply as $(x_n)_n$.

Example 1.2.2 (Examples of Sequences)
Here are a few simple examples of sequences:

- The constant sequence: $x_n = c$ for some $c \in \mathbb{R}$.
- The alternating sequence: $x_n = (-1)^n$.
- The sequence of all even numbers: $x_n = 2n$.

Definition 1.2.3 (Boundedness of a Sequence)
A sequence $(x_n)_{n\in\mathbb{N}}$ is **bounded** if there exists a real number $M \geq 0$ such that $|x_n| \leq M$ for all $n \in \mathbb{N}$.

Remark 1.2.4 (One-sided Boundedness)
A sequence can also be bounded only from above or only from below.

Example 1.2.5 ((Un-)Bounded Sequences)
The sequence $x_n = (-1)^n$ is bounded below by -1 and above by 1.
The sequence $y_n = n$ is bounded below by 1 but unbounded above.

Definition 1.2.6 (Monotonicity of a Sequence)
A sequence $(x_n)_{n\in\mathbb{N}}$ is called:

- **(monotonically) increasing**, if for all $m, n \in \mathbb{N}$: $m > n \implies x_m \geq x_n$
- **strictly** (monotonically) increasing, if for all $m, n \in \mathbb{N}$: $m > n \implies x_m > x_n$
- **(monotonically) decreasing**, if for all $m, n \in \mathbb{N}$: $m > n \implies x_m \leq x_n$
- **strictly** (monotonically) decreasing, if for all $m, n \in \mathbb{N}$: $m > n \implies x_m < x_n$.

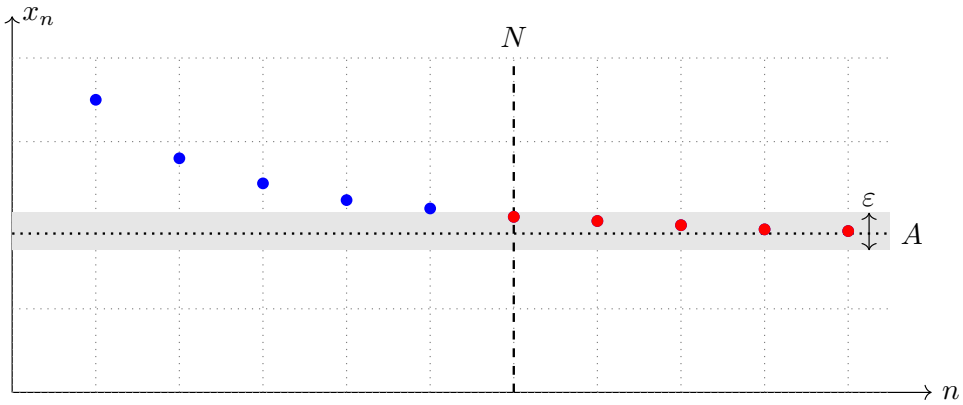
If the sequence is (strictly) monotonically increasing or decreasing, we call it (strictly) monotonic.

1.2.1 Limits

Definition 1.2.7 (Limit of a Sequence)
Let $(x_n)_{n\in\mathbb{N}}$ be a sequence. Then we denote A as the **limit** of this sequence, if $\forall \varepsilon > 0\ \exists N \in \mathbb{N}$ such that $|x_n - A| < \varepsilon\ \forall n \geq N$. One writes:

$$\lim_{n\to\infty} x_n = A$$

Remark 1.2.8 (Interpretation)
We can interpret the definition intuitively as follows:
For every arbitrarily small error margin ε, we find a large index N such that all subsequent terms are at most ε away from A. At infinity, the terms of the sequence are thus arbitrarily close to the limit A.



Remark 1.2.9
The following properties hold for the limit of a sequence:

- The limit is unique.
- The limit is well-defined under addition, multiplication, and division.

Example 1.2.10 (Typical Mistake in Applying Limit Theorems)
If two sequences converge, one can compute the product and sum of the limits individually. However, if both summands/factors do not converge individually, this is not possible!

$$1 = \lim_{n\to\infty} \frac{1}{x} \cdot x \neq \lim_{n\to\infty} \frac{1}{x} \cdot \lim_{n\to\infty} x = \text{“}0 \cdot \infty\text{“}$$

Example 1.2.11 (Limit Proof)
The sequence $x_n = \frac{1}{n}$ has limit zero.

Proof. Let $\varepsilon > 0$ be arbitrary. Then choose $N \in \mathbb{N}$ such that $N > \frac{1}{\varepsilon} > 0$. Then for all $n \geq N$

$$\left|\frac{1}{n} - 0\right| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Since $\varepsilon > 0$ was chosen arbitrarily, it follows that $\frac{1}{n}$ converges to zero. □

Remark 1.2.12 (Convergence and Boundedness)
All convergent sequences are bounded, but not all bounded sequences converge! A counterexample is the alternating sequence $x_n = (-1)^n$. It is bounded but does not converge.

Theorem 1.2.13 (Squeeze Theorem)
Let $(x_n)_{n=0}^{\infty}$, $(y_n)_{n=0}^{\infty}$ and $(z_n)_{n=0}^{\infty}$ be sequences of real numbers such that for some $N \in \mathbb{N}$ the inequalities $x_n \leq y_n \leq z_n$ hold for all $n \geq N$. Suppose that $(x_n)_{n=0}^{\infty}$ and $(z_n)_{n=0}^{\infty}$ are convergent and have the same limit. Then the sequence $(y_n)_{n=0}^{\infty}$ is also convergent, and we have:

$$\lim_{n\to\infty} x_n = \lim_{n\to\infty} y_n = \lim_{n\to\infty} z_n.$$

Remark 1.2.14 (Intuition Behind the Squeeze Theorem)
The sequences $(x_n)_{n\in\mathbb{N}}$ and $(z_n)_{n\in\mathbb{N}}$ squeeze $(y_n)_{n\in\mathbb{N}}$ tighter and tighter, so that eventually they are all ε-close to the limit, from which it follows that the sequence $(y_n)_{n\in\mathbb{N}}$ must also have this limit.

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1.2.2 Subsequences and Accumulation Points

Definition 1.2.15 (Subsequence)
Let $(x_n)_{n\in\mathbb{N}}$ be a sequence in $\mathbb{R}$. A **subsequence** of $(x_n)_{n\in\mathbb{N}}$ is a sequence of the form $(x_{n_k})_{k\in\mathbb{N}}$, where $(n_k)_{k\in\mathbb{N}}$ is a strictly monotonically increasing sequence.

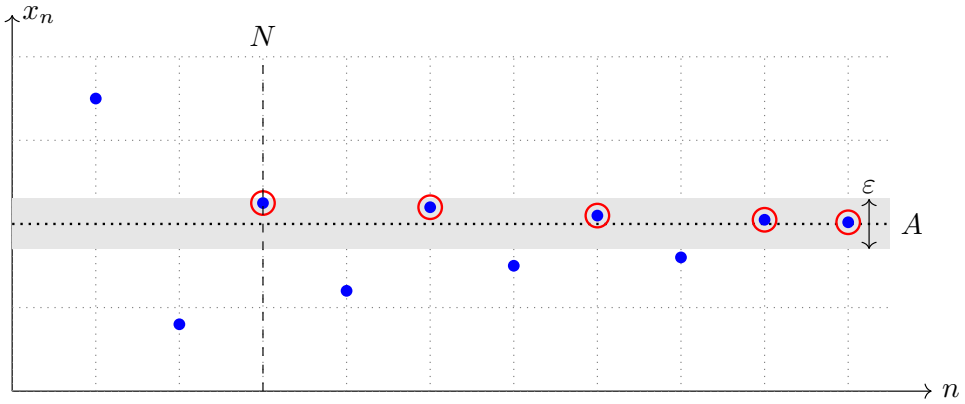
Example 1.2.16 (Simple Examples of Subsequences)
We are given the sequence of all natural numbers $x_n = n$. We want to extract only the even numbers as a subsequence $(x_{n_k})_{k\in\mathbb{N}}$. To do this, we choose $n_k = 2k$. Thus, the subsequence is $x_{n_k} = 2k$ with $k \in \mathbb{N}$.

We can extract only the positive numbers from the alternating sequence $y_n = (-1)^n$. By choosing $n_k = 2k$, we obtain $y_{n_k} = (-1)^{2k} = \left((-1)^2\right)^k = 1^k = 1$ for all $k \in \mathbb{N}$. By choosing $n_k = 2k + 1$, we obtain only the negative numbers of the sequence $(y_n)_n$.

Remark 1.2.17
From this definition it follows directly that for any subsequence we have $n_k \geq k$. Intuitively, this means that in a subsequence, terms of the original sequence $(x_n)_{n\in\mathbb{N}}$ can be omitted. But since we must always move forward, the terms of the subsequence are always keeping up with or ahead of those of the original sequence.

Definition 1.2.18 (Accumulation Point)
Let $(x_n)_{n\in\mathbb{N}}$ be a sequence in $\mathbb{R}$. A point $A \in \mathbb{R}$ is an **accumulation point** of the sequence $(x_n)_{n\in\mathbb{N}}$ if for all $\varepsilon > 0$ and every $N \in \mathbb{N}$ there exists an $n \geq N$ such that $|x_n - A| < \varepsilon$.

Remark 1.2.19
The direct difference to the simple limit of a sequence is the following: For a limit, beyond some index N, *all* subsequent terms are at most ε away. For an accumulation point, for every arbitrarily large chosen index N, there must exist at least one subsequent term that is at most ε away from the accumulation point.



Theorem 1.2.20 (Accumulation Points and Subsequences)
Let $(x_n)_{n\in\mathbb{N}}$ be a sequence. A point A is an accumulation point of $(x_n)_{n\in\mathbb{N}}$ if and only if there exists a subsequence of $(x_n)_{n\in\mathbb{N}}$ that converges to A.

Remark 1.2.21
Intuitively, an accumulation point can be thought of as a point such that infinitely many terms of the sequence are arbitrarily close. So no matter how small I choose an ε-neighborhood around my accumulation point, I can always find infinitely many terms of the sequence in this ε-neighborhood. These arbitrarily close terms come precisely from the fact that there is always a convergent subsequence.

Example 1.2.22
The alternating sequence $(-1)^n$ has the accumulation points 1 and -1. We saw in Example 1.2.16 how to construct the subsequences that converge to these accumulation points.

Remark 1.2.23
A convergent sequence has exactly **one** accumulation point, which corresponds exactly to the **limit** of the sequence. Because if a sequence converges, any arbitrary subsequence also converges to this limit (see limit superior and limit inferior 1.2.32).

Theorem 1.2.24 (Bolzano-Weierstrass Theorem)
Every bounded sequence has at least one convergent subsequence.

This theorem gives us an important property of compact intervals $[a, b] \subseteq \mathbb{R}$.

Theorem 1.2.25 (Sequential Compactness)
Let $[a, b]$ be a compact interval and $(x_n)_{n\in\mathbb{N}}$ a sequence contained in $[a, b]$. Then there exists a subsequence (x_{n_k}) such that

$$\lim_{k\to\infty} x_{n_k} = \bar{x} \quad \text{with} \quad \bar{x} \in [a, b].$$

Remark 1.2.26
The sequence $(x_n)_{n\in\mathbb{N}}$ itself does not have to converge to a limit. But since the interval in which it is contained is bounded, the sequence must be bounded and we already know that every

bounded sequence has a convergent subsequence. Because the interval is closed, it contains all its limit points.

1.2.3 Limit Superior and Limit Inferior

Limits of sequences exist only if the sequence converges. However, we can already make interesting statements for much more general sequences, even if they do not converge. Namely, we want to look at the largest and the smallest accumulation point of the sequence.

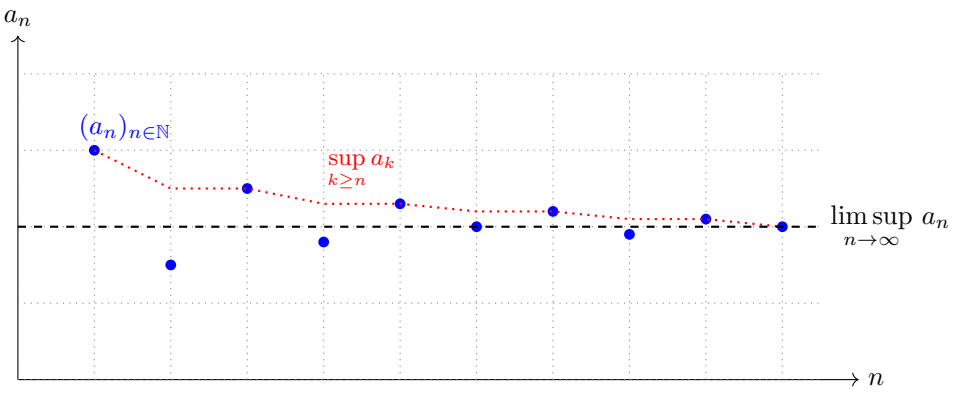
Definition 1.2.27 (Limit Superior and Limit Inferior)
Let $(x_n)_{n\in\mathbb{N}}$ be a bounded sequence of real numbers. Then the *limit superior* of $(x_n)_{n\in\mathbb{N}}$ is defined as

$$\limsup_{n\to\infty} x_n := \lim_{n\to\infty} \Big(\sup\{x_k | k \geq n\} \Big)$$

Analogously, the *limit inferior* of $(x_n)_{n\in\mathbb{N}}$ is defined as

$$\liminf_{n\to\infty} x_n := \lim_{n\to\infty} \Big(\inf\{x_k | k \geq n\} \Big).$$

Remark 1.2.28 (Intuitive Description of lim sup)
The limit superior (lim sup) of a sequence $(a_n)_{n\in\mathbb{N}}$ describes the largest accumulation value that the sequence approaches infinitely often. Formally, it is the limit of the sequence of suprema of all tail terms from index n onwards. In the graph, the terms a_n are shown as blue points. The red line connects the suprema of the tail sequences from each index n, which converge to the lim sup (black dashed line). The lim sup thus corresponds to the smallest upper bound of these suprema.



Remark 1.2.29 (Largest and Smallest Accumulation Point of a Sequence)
The lim sup is the **largest accumulation point** and the lim inf is the **smallest accumulation point**. The following inequality holds:

$$\liminf_{n\to\infty} x_n \leq x_n \leq \limsup_{n\to\infty} x_n$$

Example 1.2.30 (lim sup and lim inf)
We consider two convergent sequences a_n and b_n with limits A and B respectively. We calculate the lim sup and lim inf of the sequence

$$c_n = a_n + (-1)^n b_n.$$

We know that a_n gets closer and closer to the limit A. The term b_n will increasingly approach B. The terms of c_n will therefore eventually simply oscillate back and forth between $A+B$ and $A-B$. Thus,

$$\limsup c_n = A + B \qquad \text{and} \qquad \liminf c_n = A - B.$$

Theorem 1.2.31 (Convergence and Limit Superior / Limit Inferior)
A bounded sequence $(x_n)_{n\in\mathbb{N}}$ converges if and only if lim sup and lim inf are equal.

Remark 1.2.32 (Convergence and lim sup / lim inf)
We have already explained in 1.2.23 that if a sequence converges, every subsequence converges to the same limit. Therefore also the subsequences that have the smallest and largest accumulation point as their limit. If the sequence has a limit, then

$$\lim_{n\to\infty} x_n = \limsup_{n\to\infty} x_n = \liminf_{n\to\infty} x_n.$$

1.2.4 Cauchy Sequences

Definition 1.2.33 (Cauchy Sequence)
Let $(x_n)_{n\in\mathbb{N}}$ be a sequence in $\mathbb{R}$. We call this sequence a **Cauchy sequence** if for every $\varepsilon > 0$ there exists an $N \in \mathbb{N}$ such that $|x_n - x_m| < \varepsilon$ for all $n, m \geq N$.

A Cauchy sequence is thus a sequence where any two terms beyond some index N are at most ε apart. *All* terms get closer and closer to each other.

1 **Example 1.2.34** (Cauchy Sequence)
The sequence $x_n = \frac{1}{n}$ is a Cauchy sequence.

Proof. We choose $\varepsilon > 0$ arbitrarily and $N > \frac{1}{\varepsilon}$. Then for all $m, n \geq N$ we have

$$|x_n - x_m| = \left|\frac{1}{n} - \frac{1}{m}\right| = \left|\frac{m-n}{nm}\right| \leq \frac{|m|}{|mn|} = \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Since $\varepsilon > 0$ was chosen arbitrarily, $(x_n)_{n\in\mathbb{N}}$ is a Cauchy sequence. □

Theorem 1.2.35 (Cauchy Sequences are Bounded)
Let $(x_n)_{n\in\mathbb{N}}$ be a Cauchy sequence. Then $(x_n)_{n\in\mathbb{N}}$ is bounded.

Remark 1.2.36
The converse, that every bounded sequence is a Cauchy sequence, is of course **not true**!

Theorem 1.2.37 (Cauchy Sequences Converge (Exam))
A sequence $(x_n)_{n\in\mathbb{N}}$ converges if and only if $(x_n)_{n\in\mathbb{N}}$ is a Cauchy sequence.

Proof.

$\Longrightarrow$ Let the sequence $(x_n)_{n\in\mathbb{N}}$ converge to a limit A. Fix $\varepsilon > 0$, then there exists an $N \in \mathbb{N}$ such that $|x_n - A| < \frac{\varepsilon}{2}$ for all $n \geq N$. Let x_n and x_m be terms such that $m, n \geq N$.

$$|x_n - x_m| = |x_n - A - x_m + A| \leq |x_n - A| + |A - x_m| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \qquad \forall n, m \geq N$$

Thus, $(x_n)_{n\in\mathbb{N}}$ is a Cauchy sequence.

$\Longleftarrow$ Let $(x)_n$ be a Cauchy sequence. So there exists an N_0 such that $|x_n - x_m| < \frac{\varepsilon}{2}\ \forall n, m \geq N_0$.

We already know that every Cauchy sequence is bounded (see 1.2.4). Therefore, by the Bolzano-Weierstrass theorem, there exists a convergent subsequence $(x)_{n_k}$. The subsequence converges to A, so there exists an N_1 such that $|x_{n_k} - A| < \frac{\varepsilon}{2}\ \forall k \geq N_1$.

Choosing $N = \max\{N_0, N_1\}$ ensures that both inequalities hold. Because the index of the subsequence n_k is monotonically increasing, we have $n_N \geq N$. Therefore:

$$|x_n - A| \leq |x_n - x_{n_N}| + |x_{n_N} - A| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \qquad \forall n \geq N$$

□

Remark 1.2.38 (Convergence vs. Cauchy)
As we have seen, in $\mathbb{R}$ the convergent sequences and the Cauchy sequences are the same. This is exactly equivalent to the completeness of $\mathbb{R}$. In the definition of a Cauchy sequence, the limit is not mentioned because a priori it does not have to exist. The completeness of $\mathbb{R}$ (there are no gaps) guarantees that such a sequence has a limit. This will be explored further in Analysis II.

Multiple Choice Question 1.2.39 (FS2025, L. Kobel-Keller)
Let $(x_n)_{n=0}^{\infty}$ be a monotonic sequence. Which of the following statements is true?

(A) The sequence may have no accumulation point.

(B) The sequence has at least one accumulation point.

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(C) The sequence has exactly one accumulation point.

(D) The sequence has at least two accumulation points.

Solution:

(A) True, consider $x_n = n$ for $n \in \mathbb{N}$.

(B) False, consider $x_n = n$ for $n \in \mathbb{N}$.

(C) False, consider $x_n = n$ for $n \in \mathbb{N}$.

(D) False, consider $x_n = n$ for $n \in \mathbb{N}$.

Multiple Choice Question 1.2.40 (FS2024, A. Figalli)
Let $(x_n)_{n=0}^{\infty}$ be a sequence. Which of the following statements is not true?

(A) If $(x_n)_{n=0}^{\infty}$ is bounded, then the lim sup is an accumulation point.

(B) If $(x_n)_{n=0}^{\infty}$ is a Cauchy sequence, then the sequence is monotonic.

(C) If $(x_n)_{n=0}^{\infty}$ is bounded, then there exists a convergent subsequence.

(D) If the sequence $(x_n)_{n=0}^{\infty}$ has no accumulation points, then it is not bounded.

Solution:

(A) True. For bounded sequences, the lim sup is finite and thus well-defined. If it exists, it is always an accumulation point.

(B) False. Counterexample $x_n = (-1)^n/n$.

(C) True. Bolzano-Weierstrass 1.2.24.

(D) True. Contraposition of Bolzano-Weierstrass 1.2.24 and Theorem 1.2.20.

Multiple Choice Question 1.2.41 (HS2023, A. Figalli)
Which of the following statements is correct?

(A) Every Cauchy sequence is monotonic.

(B) Every monotonic sequence is bounded.

(C) Every bounded and monotonic sequence is a Cauchy sequence.

(D) Every bounded sequence converges.

Solution:

(A) False. Counterexample $x_n = (-1)^n/n$.

(B) False. Counterexample $x_n = n$.

(C) True. Bounded and monotonic $\implies$ convergent $\iff$ Cauchy.

(D) False. Counterexample $x_n = (-1)^n$.

Multiple Choice Question 1.2.42 (FS2024, A. Figalli)
Which of these sequences does not have a finite limit?

(A) $x_n = 2\sqrt{n+1} - \sqrt{4n+1}$

(B) $x_n = \sqrt{2n+1} - \sqrt{n}$

(C) $x_n = \sqrt[3]{n^3+n^2} - n$

(D) $x_n = \sqrt{n^2+n} - \sqrt{n^2+2}$

Solution:

(A) False. We use the conjugate expression

$$\sqrt{4n+4} - \sqrt{4n+1} = \frac{(4n+4)-(4n+1)}{\sqrt{4n+4}+\sqrt{4n+1}} = \frac{3}{\sqrt{4n+4}+\sqrt{4n+1}} \xrightarrow{n\to\infty} 0$$

(B) True. We use the conjugate expression

$$\sqrt{2n+1} - \sqrt{n} = \frac{2n+1-n}{\sqrt{2n+1}+\sqrt{n}} = \frac{n+1}{\sqrt{2n+1}+\sqrt{n}} \xrightarrow{n\to\infty} \infty$$

(C) False. We use the identity $a^3 - b^3 = (a-b)(a^2+ab+b^2)$ and write

$$\sqrt[3]{n^3+n^2} - n = \frac{n^3+n^2-n^3}{(n^3+n^2)^{2/3} + n(n^3+n^2)^{1/3} + n^2} = \frac{n^2}{n^2\big((1+\frac{1}{n})^{2/3} + (1+\frac{1}{n})^{1/3} + 1\big)} \xrightarrow{n\to\infty} \frac{1}{3}.$$

We will learn better methods for calculating such limits later in the semester (Taylor expansion and L'Hôpital's rule).

(D) False. We use the conjugate expression

$$\sqrt{n^2+n}-\sqrt{n^2+2}=\frac{(n^2+n)-(n^2+2)}{\sqrt{n^2+n}+\sqrt{n^2+2}}=\frac{n-2}{n\left(\sqrt{1+\frac{1}{n}}+\sqrt{1+\frac{2}{n}}\right)}\xrightarrow{n\to\infty}\frac{1}{2}.$$

1.3 Sets

1.3.1 Openness and Closedness, Intervals

Definition 1.3.1 (Interval)
We define the closed interval $[a,b]$ as

$$[a,b]=\{x\in\mathbb{R}|a\leq x\leq b\}$$

the open interval (a,b) as

$$(a,b)=\{x\in\mathbb{R}|a<x<b\}$$

From these, half-open intervals can be derived. Intervals in $\mathbb{R}$ are *connected*.

Connected sets will be discussed in Analysis II; here they are only mentioned.

Definition 1.3.2 (Open and Closed Set)
A subset $U\subseteq\mathbb{R}$ is called **open** in $\mathbb{R}$, if for every $x\in U$ there exists an open interval I such that $x\in I$ and $I\subseteq U$.
A set $F\subseteq\mathbb{R}$ is called **closed**, if its complement $\mathbb{R}\setminus F$ is open.

1 Intuitively, openness means that for every point in my set, I can always find more points in the set arbitrarily close.

Remark 1.3.3
„Not open" does not directly mean „closed" and vice versa. There are sets that are neither open nor closed. For example, half-open intervals $[0,1)$.

Remark 1.3.4 (Properties of Open and Closed Sets)
The following properties hold for open and closed sets:

- Arbitrary unions of open sets are open.
- Finite intersections of open sets are open.
- Arbitrary intersections of closed sets are closed.
- Finite unions of closed sets are closed.

Multiple Choice Question 1.3.5 (FS2024, A. Figalli)
Are the following sets in $\mathbb{R}$ open, closed, both, or neither?

(A) $\{2\}\cup(2,3]$

(B) $\{0\}\cup(1,\infty)$

(C) $\left\{\frac{1}{n}:n\in\mathbb{N},n\geq 1\right\}$

(D) $[0,1)\cup(1,2]$

Solution:

(A) This set is closed. It corresponds to the interval $[2, 3]$.

(B) This set is neither open nor closed.

(C) This set is neither open nor closed.

(D) This set is neither open nor closed.

Multiple Choice Question 1.3.6
Are the following sets in $\mathbb{R}$ open, closed, both, or neither?

(A) $\{0\}$

(B) $(0, \infty)$

(C) $\{0\} \cup \left\{\frac{1}{n} : n \in \mathbb{N}, n \geq 1\right\}$

(D) $(-1, 1) \cup \{2\}$

Solution:

(A) This set is closed.

(B) This set is open.

(C) This set is closed.

(D) This set is neither open nor closed.

1

1.3.2 Maximum and Supremum

Definition 1.3.7 (Bounded Set, Maximum, Minimum)
Let $X \subseteq \mathbb{R}$ be a subset of the real numbers.

- X is bounded from above, if there exists a real number $s \in \mathbb{R}$ such that $x \leq s$ for all $x \in X$. Any such number s is called an **upper bound** of X. An upper bound that is itself contained in X is called a **maximum**.

$$s = \max(X)$$

- **Bounded from below**, **lower bound**, and **minimum** are defined completely analogously.
- $X \subseteq \mathbb{R}$ is **bounded** if it is bounded from above and below.

Remark 1.3.8
If a maximum (or minimum) exists for a set, then it is unique!

Definition 1.3.9 (Supremum (Infimum follows analogously))
Let $X \subseteq \mathbb{R}$ be a subset and let $A := \{a \in \mathbb{R} | x \leq a \ \forall x \in X\}$ be the set of all upper bounds of X. If a minimum $s = \min(A)$ exists, then we call it the **supremum of** X and write

$$s = \sup(X)$$

The supremum is always well-defined for a subset bounded from above, because the subset of upper bounds is therefore always bounded from below and closed. Hence the minimum of A exists and therefore the supremum of X exists.

Remark 1.3.10
If the maximum exists, then the maximum and the supremum **always** coincide.

The infimum of a subset can be defined completely analogously.

Multiple Choice Question 1.3.11 (HS2022, G. Felder)
The supremum of a non-empty bounded subset A of $\mathbb{R}$ is

(A) the smallest number $s \in \mathbb{R}$ such that $x \leq s$ for all $x \in A$.

(B) the smallest number $s \in A$ such that $x \leq s$ for all $x \in A$.

(C) the smallest number $s \in A$ such that there exists an $x \in A$ with $x \leq s$.

(D) the smallest number $s \in \mathbb{R}$ such that there exists an $x \in A$ with $x \leq s$.

Solution:(A)

Multiple Choice Question 1.3.12 (HS2025, L. Kobel-Keller)
We consider the set

$$A := \bigcup_{n \geq 1} A_n \qquad \text{with} \quad A_n := \left[\frac{1}{n}, 1 + \frac{1}{n}\right), n \geq 1.$$

1 Then:

(A) $\inf A = 0$ and $\max A = \sup A = 2$, the minimum of A is not attained.

(B) $\inf A = 0$ and $\sup A = 2$, neither the minimum nor the maximum of A is attained.

(C) $\min A = \inf A = 0$ and $\max A = \sup A = 2$.

(D) $\min A = \inf A = 0$ and $\sup A = 2$, the maximum of A is not attained.

Solution:(B)

1.3.3 Compactness

Definition 1.3.13 (Compact Subsets)
A subset $U \subseteq \mathbb{R}$ is **compact** if and only if it is closed and bounded.

Remark 1.3.14
The **compact intervals** in $\mathbb{R}$ are exactly the closed intervals $[a, b]$.

Example 1.3.15
Compact sets in $\mathbb{R}$ are for example:

$$[0, 1] \qquad \{1\} \qquad [0, 1] \cup [2, 3]$$

There are many different (but completely equivalent) characterizations of compactness (see Analysis II). In Analysis I, however, sequential compactness (see 1.3.16) is the most common definition.

1

Theorem 1.3.16 (Compact Subsets)
A subset $[a, b] \subseteq \mathbb{R}$ is **compact** if and only if every sequence in $[a, b]$ has a convergent subsequence (with limit in $[a, b]$!).

Chapter 2

Functions in One Real Variable

2.1 Monotonicity and Boundedness

Definition 2.1.1 (Monotonic Function)
Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$. The function f is called

- **(monotonically) increasing** if for all $x, y \in D$, we have

$$x < y \implies f(x) \leq f(y),$$

- **strictly (monotonically) increasing** if for all $x, y \in D$, we have

$$x < y \implies f(x) < f(y),$$

Analogously, **(monotonically) decreasing** and **strictly (monotonically) decreasing** can be defined.

Definition 2.1.2 (Bounded Function)
Let $D \subseteq \mathbb{R}$ be a non-empty subset and $f : D \to \mathbb{R}$ a function. The function f is **bounded from above** if there exists a constant $M > 0$ such that

$$f(x) \leq M \text{ for all } x \in D.$$

The function f is **bounded from below** if there exists a constant $M > 0$ such that

$$f(x) \geq -M \text{ for all } x \in D.$$

The function f is **bounded** if it is bounded from above and below, i.e., $|f(x)| \leq M$.

2.2 Characterizations of Continuity

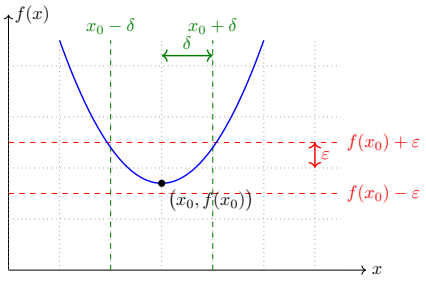
Definition 2.2.1 (Continuity at a point x_0)
Let $D \subseteq \mathbb{R}$ be a subset and $f : D \to \mathbb{R}$ a function. We say that f is **continuous at the point** $x_0 \in D$ if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$\forall x \in D, \qquad |x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon.$$

The function f is **continuous** on D if f is continuous at every point in D.

Remark 2.2.2 (Intuition for Continuity)
Continuity at the point x can also be interpreted using intervals. It means that for every arbitrarily small ε-neighborhood of $f(x)$, there exists a δ-neighborhood whose image under f is contained in the ε-neighborhood. Explicitly, $\forall \varepsilon > 0\ \exists \delta > 0$ such that

$$f((x-\delta, x+\delta)) \subseteq (f(x)-\varepsilon, f(x)+\varepsilon).$$



Remark 2.2.3 (Types of Discontinuities)
There are the following types of points of discontinuity:

- Removable Discontinuity
  Here both one-sided limits exist and are equal, but are not equal to the function value.
- Jump Point
  Here both one-sided limits exist, but are not equal.
- Poles and Infinite Oscillations
  All other cases, i.e., where at least one of the one-sided limits does not exist – is infinite (pole) or oscillates infinitely.

Definition 2.2.4 (Uniform Continuity)
Let $D \subseteq \mathbb{R}$ be a subset and $f : D \to \mathbb{R}$ a function. We say that f is *uniformly continuous* if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$$

for all $x, y \in D$.

Remark 2.2.5
In „ordinary" continuity, we fix the δ with the existential quantifier *after* we have chosen our point x_0. Therefore, our δ a priori depends on x_0, and there does not need to be a *universal* δ that works for all $x \in D$ for a given ε. *Uniform continuity* now demands exactly that. So fix an ε, then choose a corresponding δ, but this must work for all $x \in D$. Here, δ may no longer depend on the point at which I check continuity!

Example 2.2.6 (Continuous vs. Uniformly Continuous)
Briefly explain why a uniformly continuous function is always continuous, but a continuous function is not always uniformly continuous.

Solution: A uniformly continuous function is always continuous because uniform continuity is a special case where δ does not depend on the point.

However, a continuous function is not always uniformly continuous because a priori δ can indeed depend on our point. This does not disturb continuity because for each point in the domain, its own δ can be chosen. But uniform continuity requires a δ independent of the point, which does not necessarily exist for every continuous function.

It is very tedious to carry out continuity proofs using the ε-δ criterion. Therefore, it is helpful to use an equivalent criterion, namely the so-called **sequential continuity**.

Theorem 2.2.7 (Continuity $\iff$ Sequential Continuity (Exam))
Let $D \subseteq \mathbb{R}$ be a set, let $f : D \to \mathbb{R}$ be a function, and $x_0 \in D$. The function f is continuous at the point x_0 if and only if, for all sequences $(x_n)_n \subseteq D$ converging to x_0, $f((x_n)_n)$ converges to $f(x_0)$. That is:

$$\lim_{n\to\infty} f(x_n) = f(x_0) = f\left(\lim_{n\to\infty} x_n\right).$$

Proof.

$\implies$ Let $(x_n) \to x_0$ be a sequence in D and f a continuous function. Fix an arbitrary $\varepsilon > 0$, by the continuity of f there exists a corresponding $\delta > 0$. Since our sequence (x_n) converges to x_0, by definition there exists an $N \in \mathbb{N}$ such that $|x_n - x_0| < \delta \ \forall n \geq N$. Therefore, by the continuity of f, we have $|f(x_n) - f(x_0)| < \varepsilon \ \forall n \geq N$. Thus we have

$$\lim_{n\to\infty} f(x_n) = f(x_0)$$

and hence f is sequentially continuous.

$\impliedby$ We show this implication via contraposition. Suppose f is not continuous. Then there exists an $\varepsilon > 0$ such that for all $\delta > 0$ we have $|x_n - x_0| < \delta$, but $|f(x_n) - f(x_0)| \geq \varepsilon$. Choosing $\delta = 2^{-n}$, we have $x_n \to x_0$ in D, but our $f(x_n)$ does not necessarily converge to $f(x_0)$. Thus f is not sequentially continuous.

□

1

Remark 2.2.8 (Recursively Defined Sequences)
If a recursively defined sequence converges to a limit A, then this limit is a *fixed point* of the recurrence equation. The idea is that the recurrence rule can be viewed as a continuous function f, with $x_n = f(x_{n-1})$. Then by sequential continuity, we have

$$\lim_{n\to\infty} x_n = \lim_{n\to\infty} f(x_{n-1}) = f\left(\lim_{n\to\infty} x_{n-1}\right) \iff A = f(A).$$

Thus, the limit, *if it exists*, must be a fixed point of the recurrence equation.

2.3 Consequences of Continuity

2.3.1 Intermediate Value Theorem

Theorem 2.3.1 (Intermediate Value Theorem (Exam))
Let $f : [a, b] \to \mathbb{R}$ be a continuous function with $f(a) \leq f(b)$. Then for every real number c with $f(a) \leq c \leq f(b)$, there exists $\bar{x} \in [a, b]$ such that $f(\bar{x}) = c$.

Proof. The argument works by showing that $\begin{cases} f(\bar{x}) \leq c \\ f(\bar{x}) \not< c \end{cases} \implies f(\bar{x}) = c.$
We obtain the first relation via the continuity of f. Let

$$X = \{x \in [a, b] \mid f(x) \leq c\}$$

Because X is bounded ($X \subseteq [a, b]$) and non-empty ($a \in X$), a supremum exists and $\sup(X) = \bar{x}$. Construct a sequence $(x)_n^\infty \to \bar{x} \subseteq X$ as $x_n \in X \cap [\bar{x} - 2^{-n}, \bar{x}]$. Such a sequence is well-defined because the intersection is never empty. Then by the continuity of f, we have

$$f(\bar{x}) = \lim_{n\to\infty} f(x_n) \leq c,$$

since $f(x_n) \leq c$, because $(x_n)_n^\infty \subseteq X$.

We show the second relation by contradiction. Suppose $f(\bar{x}) < c$. Choose $\varepsilon = c - f(\bar{x}) > 0$. Due to the continuity of f, we have:

$$|x - \bar{x}| < \delta \implies |f(x) - f(\bar{x})| < \varepsilon = c - f(\bar{x}).$$

Thus $f(x) - f(\bar{x}) < \varepsilon = c - f(\bar{x}) \implies f(x) < c$. Therefore, we find function values in the interval $(\bar{x}, \bar{x} + \delta)$ with $f(x) < c$. This contradicts the definition of $\bar{x}$ as $\sup(X)$.

Hence $f(\bar{x}) = c$ and the theorem follows.

□

Example 2.3.2 (Intermediate Value Property $\implies$ Continuity)
Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be a monotonic function. For all $c \in [f(a), f(b)]$ there exists an $x \in [a, b]$ with $f(x) = c$. Show that f is continuous on the interval $[a, b]$.

Proof. Without loss of generality, assume f is monotonically increasing. Otherwise consider $-f$.

We prove by contraposition. So assume f is not continuous. Then there is a jump discontinuity x_0 in our interval (the only possibility because f is monotonic), where we have:

1

$$\lim_{x\to x_0^-} f(x) < \lim_{x\to x_0^+} f(x).$$

Since f is monotonically increasing, there is no way for the function values in between to be attained. So for these function values in the interval $\left(\lim_{x\to x_0^-} f(x), \lim_{x\to x_0^+} f(x)\right)$ there is no $x \in [a, b]$ with $f(x) \in \left(\lim_{x\to x_0^-} f(x), \lim_{x\to x_0^+} f(x)\right)$. Therefore, the function does not satisfy our intermediate value property and the theorem follows. □

2.3.2 Inverse Function Theorem

Theorem 2.3.3 (Inverse Function Theorem)
Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ a continuous, strictly monotonic function. Then $f(I) \subseteq \mathbb{R}$ is again an interval and the function $f : I \to f(I)$ has a continuous, strictly monotonic inverse function $f^{-1} : f(I) \to I$.

Remark 2.3.4
It is important to recognize that a continuous, strictly monotonic function on an interval is always injective. Restricting the codomain to $f(I)$ thus gives us a bijective function, for which an inverse always exists. This is the essence of the Inverse Function Theorem.

2.3.3 Continuous Functions on Compact Intervals

Recall, a set in $\mathbb{R}$ is compact if it is **closed** and **bounded**. Compact intervals are therefore precisely the closed intervals $[a, b]$ with $a < b$.

Theorem 2.3.5 (Compact Intervals and Uniform Continuity (Exam))
Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Then f is also uniformly continuous on $[a, b]$.

Proof. Assume, by contradiction, that f is not uniformly continuous on $[a, b]$. Then there exists $\varepsilon > 0$ such that for every $\delta > 0$ one can find $x, y \in [a, b]$ with

$$|x - y| < \delta \quad \text{and} \quad |f(x) - f(y)| \geq \varepsilon.$$

Taking $\delta = 2^{-n}$ for each $n \in \mathbb{N}$, we obtain sequences $(x_n)_{n=0}^{\infty}$ and $(y_n)_{n=0}^{\infty}$ in $[a, b]$ with

$$|x_n - y_n| < 2^{-n} \quad \text{and} \quad |f(x_n) - f(y_n)| \geq \varepsilon.$$

Since $[a, b]$ is compact, the sequence $(x_n)_{n=0}^{\infty}$ has a convergent subsequence $(x_{n_k})_{k=0}^{\infty}$ with limit $\bar{x} \in [a, b]$. Then

$$|y_{n_k} - \bar{x}| \leq |y_{n_k} - x_{n_k}| + |x_{n_k} - \bar{x}| < 2^{-n_k} + |x_{n_k} - \bar{x}| \underset{k\to\infty}{\longrightarrow} 0,$$

so $y_{n_k} \to \bar{x}$ as well. By the continuity of f, the images of these sequences converge to $f(\bar{x})$:

$$\lim_{k\to\infty} f(x_{n_k}) = \lim_{k\to\infty} f(y_{n_k}) = f(\bar{x}).$$

Therefore

$$|f(x_{n_k}) - f(y_{n_k})| \leq |f(x_{n_k}) - f(\bar{x})| + |f(\bar{x}) - f(y_{n_k})| \underset{k\to\infty}{\longrightarrow} 0,$$

which contradicts our assumption. Hence, f is uniformly continuous on $[a, b]$. □

Definition 2.3.6 (Maxima and Minima of Functions)
Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$ be a real-valued function on D. We say that the function f attains its maximum value at a point $x_0 \in D$ if $f(x) \leq f(x_0)$ for all $x \in D$. We call $f(x_0)$ the **maximum** of f. Analogously, f attains its minimum value at $x_0 \in D$ if $f(x) \geq f(x_0)$ for all $x \in D$. In this case we call $f(x_0)$ the **minimum** of f. Maxima and minima are collectively referred to as **extreme values** or **extrema**.

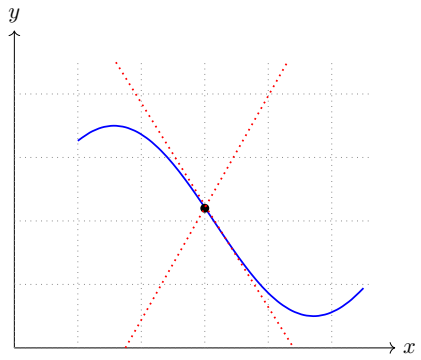
Theorem 2.3.7 (Continuous Functions on Compact Intervals)
Let $[a, b]$ be a compact interval and $f : [a, b] \to \mathbb{R}$ a continuous function. Then f attains both a maximum and a minimum.

2.4 Lipschitz Continuity

Definition 2.4.1 (Lipschitz Continuity)
Let $D \subseteq \mathbb{R}$ be a subset. A function $f : D \to \mathbb{R}$ is called **Lipschitz continuous** if there exists an $L \geq 0$ such that

$$|f(x) - f(y)| \leq L|x - y|$$

for all $x, y \in D$.



Remark 2.4.2 (Intuitive Explanation of Lipschitz Continuity)
Lipschitz continuity intuitively means that one can find a cone (red dotted) with slope L such that no function values lie inside the cone, no matter where it is placed on the function. The value L is thus independent of the point, just like in uniform continuity.

Remark 2.4.3
It is important to remember the strength of the different continuity concepts. The following chain of implications holds:

$$\text{Lipschitz continuous} \implies \text{Uniformly continuous} \implies \text{Continuous}$$

Example 2.4.4 (Lipschitz $\implies$ Uniformly continuous)
Show that a Lipschitz continuous function f is uniformly continuous. Give an example of a uniformly continuous function that is not Lipschitz continuous.

Solution: Fix an $\varepsilon > 0$. Choose our $\delta = \frac{\varepsilon}{L}$, where L is the Lipschitz constant. We have $|x - \bar{x}| < \delta$ and we can estimate

$$|f(x) - f(\bar{x})| \leq L|x - \bar{x}| < L\delta = L \cdot \frac{\varepsilon}{L} = \varepsilon.$$

1 We see that δ does not depend on $\bar{x}$! Because $\bar{x}$ in our domain and $\varepsilon > 0$ were chosen arbitrarily, f is uniformly continuous.

The function $f(x) = \sqrt{x}$ on $[0, \infty)$ is uniformly continuous, but not Lipschitz continuous.

2.5 Sequences of Functions

Now we consider sequences of functions. An example is $f_n(x) = x^n$ on the interval $[0, 1]$. We now ask what the expression $\lim_{n\to\infty} f_n(x)$ means and what the resulting function is. There are two ways to characterize the convergence of sequences of functions.

2.5.1 Pointwise Convergence

Definition 2.5.1 (Pointwise Convergence of Sequences of Functions)
Let $D \subseteq \mathbb{R}$, let (f_n) be a sequence of functions $f_n : D \to \mathbb{R}$, and let $f : D \to \mathbb{R}$ be a function. We say that the sequence (f_n) **converges pointwise to** f if for each $x \in D$ the sequence $\big(f_n(x)\big)$ converges to $f(x)$. In this case we call f the **pointwise limit** of (f_n).

Remark 2.5.2
This definition, however, has a fundamental problem. The properties of f, such as continuity, are not preserved under this convergence. A sequence of continuous functions can have a non-continuous limit. Like our example $f_n(x) = x^n$, which converges to

$$f(x) = \begin{cases} 0 & x < 1 \\ 1 & x = 1 \end{cases}.$$

2.5.2 Uniform Convergence

Uniform convergence is a sufficiently strong convergence criterion that preserves the continuity of the sequence of functions.

Definition 2.5.3 (Uniform Convergence of Sequences of Functions)
Let $D \subseteq \mathbb{R}$, let (f_n) be a sequence of functions $f_n : D \to \mathbb{R}$, and let $f : D \to \mathbb{R}$ be a function. We say that the sequence (f_n) **converges uniformly to** f if for every $\varepsilon > 0$ there exists an $N \in \mathbb{N}$ such that for all $n \geq N$ and all $x \in D$ we have

$$|f_n(x) - f(x)| < \varepsilon.$$

Remark 2.5.4
Uniform convergence can also be equivalently formulated as

$$\lim_{n\to\infty} \left(\sup_{x\in D} |f_n(x) - f(x)| \right) = 0.$$

One can interpret it as: for all $n \in N$ we choose, via the supremum, the maximum deviation of f_n and f. If this maximum deviation converges to zero, then f_n converges uniformly to f. Caution, this is not a lim sup (limit superior)!

Example 2.5.5 (Counterexample to Uniform Convergence)
We prove that the sequence of functions

$$f_n : [0,1] \mapsto \mathbb{R} \qquad f_n(x) = x^n$$

does not converge uniformly on $[0,1]$.

Proof. We first find the pointwise limit. For this we find

$$f(x) = \begin{cases} 0 & x \in [0,1) \\ 1 & x = 1 \end{cases}.$$

Now we first compute

$$\sup_{x\in[0,1]} |x^n - f(x)|$$

and distinguish the two cases $x \in [0,1)$ and $x = 1$. We find

$$\sup_{x\in[0,1]} |x^n - f(x)| = \begin{cases} \sup\limits_{x\in[0,1)} |x^n - 0| = 1 \\ \sup\limits_{x=1} |x^n - 1| = 0 \end{cases} = 1.$$

Thus the supremum does not depend on n at all and the limit is therefore also 1. We have

$$\lim_{n\to\infty} \sup_{x\in[0,1]} |x^n - f(x)| = 1 \neq 0.$$

Hence f_n does not converge uniformly to f.

□

Theorem 2.5.6 (Uniform Convergence Preserves Continuity)
Let (f_n) be a sequence of continuous functions $f_n : D \subseteq \mathbb{R} \to \mathbb{R}$ that converges uniformly to a function $f : D \to \mathbb{R}$. Then f is continuous.

Proof. We fix an $\varepsilon > 0$ and a $\bar{x} \in D$. Since $(f_n)_{n\in\mathbb{N}}$ converges uniformly to f, we can find an $N \in \mathbb{N}$ such that $|f_n(x) - f(x)| < \frac{\varepsilon}{3}$. Furthermore, due to the continuity of f_n, for all $|x - \bar{x}| < \delta$, we have $|f_n(x) - f_n(\bar{x})| < \frac{\varepsilon}{3}$. Now we estimate

$$|f(x) - f(\bar{x})| \leq |f(x) - f_n(x)| + |f_n(x) - f(\bar{x})| \leq |f(x) - f_n(x)| + |f_n(x) - f_n(\bar{x})| + |f_n(\bar{x}) - f(\bar{x})|$$

Thus for all $|x - \bar{x}| < \delta$, we have $|f(x) - f(\bar{x})| < 3 \cdot \frac{\varepsilon}{3} = \varepsilon$. Because ε and $\bar{x} \in D$ were chosen arbitrarily, $f(x)$ is continuous on D. □

Remark 2.5.7
With this theorem, we can abbreviate Example 2.5.5. We see that $f_n(x) = x^n$ is continuous, but the pointwise limit $f(x)$ is not. Hence f_n cannot converge uniformly to f.

Multiple Choice Question 2.5.8 (HS2023, A. Figalli)
Let I be a bounded interval and $f : I \to \mathbb{R}$ a continuous function. Which of the following statements could be false?

(A) If f is injective, then f is monotonic.

(B) $f(I)$ is open.

(C) (Optional) If f is bounded, then f is integrable.

(D) $f(I)$ is an interval.

Solution:

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(A) True. We know that because f is injective it has to „move" the whole time, and it cannot jump because it is continuous. Thus it is monotonic.

(B) False, we only know that the preimages of open sets are open!

(C) This is true. (See section on the Riemann Integral.)

(D) This is true, because of the intermediate value theorem.

Multiple Choice Question 2.5.9 (HS2022, G. Felder)
Let $a < b$. What holds for all functions $f : [a, b] \to \mathbb{R}$?

(A) If f is continuous, then f is differentiable.

(B) If f is monotonically increasing, then f is bounded.

(C) (Optional) If f is Riemann-integrable, then f is continuous.

(D) If f attains a maximum and a minimum, then f is continuous.

Solution:

(A) False. Take $|x|$.

(B) True. We know that $f(b)$ is the upper bound.

(C) False. Take any step function.

(D) False. Take any step function.

2.6 Landau Notation

Definition 2.6.1 (Landau Big-$\mathcal{O}$)
Let $f, g : D \to \mathbb{R}$. We write

$$f(x) = \mathcal{O}(g(x)) \text{ as } x \to x_0$$

if there exist a constant $M > 0$ and a $\delta > 0$ such that

$$x \in D \cap (x_0 - \delta, x_0 + \delta) \implies |f(x)| \leq M|g(x)|.$$

Then we say that f is a Big-$\mathcal{O}$ of g as $x \to x_0$.

Remark 2.6.2
Intuitively, $f \in \mathcal{O}(g)$ as $x \to x_0$ means that f goes to x_0 at least as fast as g. Or one can imagine that

$$\limsup_{x \to x_0} \frac{|f(x)|}{|g(x)|} \leq M.$$

So if we look sufficiently close to x_0, f is bounded by $\pm M \cdot g$ for some constant $M > 0$.

Remark 2.6.3 (Big-$\mathcal{O}$ towards ∞)
The point x_0 can also be "replaced" with $\pm\infty$ to study the limiting behavior of the two functions. However, the behavior usually reverses! For example,

$$x^2 \in \mathcal{O}(x) \quad \text{as } x \to 0 \qquad \text{but} \qquad x \in \mathcal{O}(x^2) \quad \text{as } x \to \infty.$$

Definition 2.6.4 (Landau Little-o)
Let $f, g : D \to \mathbb{R}$. We write

$$f(x) = o(g(x)) \text{ as } x \to x_0$$

if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$x \in D \cap (x_0 - \delta, x_0 + \delta) \implies |f(x)| \leq \varepsilon|g(x)|.$$

Then we say that f is a little-o of g as $x \to x_0$.

Remark 2.6.5 (Little-o vs. Big-$\mathcal{O}$)
Intuitively, little-o is exactly the same as Big-$\mathcal{O}$ but with constant $M = 0$. Thus little-o is more specific and provides a sharper bound than Big-$\mathcal{O}$.

Chapter 3

Series

3.1 Convergence of Series

Definition 3.1.1 (Convergent Series)
Let (a_n) be a sequence of real numbers, and let $A \in \mathbb{R}$. We say that the series $\sum_{k=0}^{\infty} a_k$ converges to A if

$$\lim_{n\to\infty} \sum_{k=0}^{n} a_k = \lim_{n\to\infty} s_n = A.$$

Here s_n is the sequence of **partial sums**. We call a_n the n-th term or the n-th summand of the series.

We call the series $\sum_{k=0}^{\infty} a_k$ **convergent** if s_n has a limit; in this case we call it the **value of the series**. Otherwise, we say the series is not convergent. If s_n **diverges** to ∞ (resp. to $-\infty$), we call the series $\sum_{k=0}^{\infty} a_k$ divergent to ∞ (resp. $-\infty$).

Theorem 3.1.2 (Necessary Condition for Series Convergence)
If the series $\sum_{k=0}^{\infty} a_k$ converges, then the sequence (a_k) must be a null sequence. Thus this is a necessary condition for the convergence of the series $\sum_{k=0}^{\infty} a_k$.

Remark 3.1.3 (Convergence of the Tail)
It suffices to consider the series starting from an arbitrary $N \in \mathbb{N}$. Suppose we want to investigate the behavior of the series $\sum_{k=0}^{\infty} a_k$. Then it is enough to consider the behavior of the tail of the partial sums.

$$\sum_{k=0}^{\infty} a_k = \sum_{k=0}^{N-1} a_k + \sum_{k=N}^{\infty} a_k$$

The first summand is always a finite number and therefore always converges. Thus, only the tail, i.e., the series from some arbitrarily large $N \in \mathbb{N}$ onward, is relevant to investigate.

3.1.1 Absolute and Conditional Convergence

Definition 3.1.4 (Absolute and Conditional Convergence)
We say that a series $\sum_{k=0}^{\infty} a_k$ is **absolutely convergent** if the series $\sum_{k=0}^{\infty} |a_k|$ converges.

The series $\sum_{k=0}^{\infty} a_k$ is **conditionally convergent** if it converges but is not absolutely convergent.

It turns out that only **absolutely convergent** series have truly stable convergence properties. An example is the **Riemann rearrangement theorem**.

3.1.2 Convergence Tests

Theorem 3.1.5 (Majorant and Minorant Test)
Let $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$ be two series such that $0 \leq a_k \leq b_k$ for all $k \in \mathbb{N}$. Then $0 \leq \sum_{k=0}^{\infty} a_k \leq \sum_{k=0}^{\infty} b_k$, and in particular we have

$$\sum_{k=0}^{\infty} b_k \text{ convergent} \implies \sum_{k=0}^{\infty} a_k \text{ convergent},$$

$$\sum_{k=0}^{\infty} a_k \text{ divergent to } \infty \implies \sum_{k=0}^{\infty} b_k \text{ divergent to } \infty.$$

Example 3.1.6
Using the majorant test, show that the following series converges:

$$\sum_{n=1}^{\infty} \frac{1}{n2^n}.$$

Proof. We know that for each $n \geq 1$ we have

$$\frac{1}{n2^n} \leq \frac{1}{2^n} \iff n2^n \geq 2^n \iff n \geq 1.$$

Thus, convergence follows from the majorant test

$$\sum_{n=1}^{\infty} \frac{1}{n2^n} \leq \sum_{n=1}^{\infty} \frac{1}{2^n} < \infty,$$

because $\sum_{n=1}^{\infty} \frac{1}{2^n}$ is a geometric series with $|q| = |\frac{1}{2}| < 1$. □

A useful test for transforming a series into another series with the same convergence properties is the condensation test.

Theorem 3.1.7 (Cauchy Condensation Test)
Let (a_n) be a monotonically decreasing sequence of nonnegative real numbers. Then

$$\sum_{k=0}^{\infty} a_k \text{ convergent} \iff \sum_{k=0}^{\infty} 2^k a_{2^k} \text{ convergent}.$$

Example 3.1.8
Consider the series

$$\sum_{n=1}^{\infty} \frac{1}{n^s},$$

where $s > 0$ is a real parameter. This series defines the *Riemann zeta function* $\zeta(s)$ for real $s > 1$.

Apply the *Cauchy condensation test*. Since the terms $a_n = n^{-s}$ are positive and monotonically decreasing for $s > 0$, we examine the condensed series

$$\sum_{k=0}^{\infty} 2^k a_{2^k} = \sum_{k=0}^{\infty} 2^k \cdot \frac{1}{(2^k)^s} = \sum_{k=0}^{\infty} 2^k \cdot 2^{-ks} = \sum_{k=0}^{\infty} \left(2^{(1-s)}\right)^k.$$

This is a geometric series and it converges if and only if

$$2^{1-s} < 1 \quad \Longleftrightarrow \quad 1 - s < 0 \quad \Longleftrightarrow \quad s > 1.$$

By the Cauchy condensation test, the original series $\sum_{n=1}^{\infty} n^{-s}$ converges precisely for $s > 1$ and diverges for $0 < s \leq 1$.

Theorem 3.1.9 (Leibniz Test)
Let (a_n) be a monotonically decreasing sequence of nonnegative real numbers that converges to zero. The alternating series $\sum_{k=0}^{\infty}(-1)^k a_k$ converges and

$$\sum_{k=0}^{2N+1}(-1)^k a_k \leq \sum_{k=0}^{\infty}(-1)^k a_k \leq \sum_{k=0}^{N}(-1)^k a_k$$

holds for all $N \in \mathbb{N}$.

Example 3.1.10 (Alternating Harmonic Series)
Consider the series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \cdots.$$

This is called the *alternating harmonic series*. All conditions of the Leibniz Test are met, because $1/n$ decreases monotonically to zero. Therefore, the alternating harmonic series *converges* by the Leibniz test. Note that the (non-alternating) harmonic series *diverges!*

Theorem 3.1.11 (Generalized Triangle Inequality)
An absolutely convergent series $\sum_{k=0}^{\infty} a_k$ is also convergent and satisfies the generalized triangle inequality

$$\left|\sum_{k=0}^{\infty} a_k\right| \leq \sum_{k=0}^{\infty} |a_k|.$$

Remark 3.1.12
Note here that the series must be **absolutely convergent** for this inequality to make sense. Otherwise, the right-hand side is ∞.

Theorem 3.1.13 (Cauchy Root Test (Exam))
Let $(a_n)_n$ be a real sequence and

$$\alpha = \limsup_{n\to\infty} \sqrt[n]{|a_n|} \in \mathbb{R} \cup \{\infty\}.$$

Then

$$\alpha < 1 \implies \sum_{n=0}^{\infty} a_n \qquad \text{converges absolutely}$$

and

$$\alpha > 1 \implies \sum_{n=0}^{\infty} a_n \qquad \text{does not converge.}$$

Proof. Let $\alpha < 1$ and define $q = \frac{1+\alpha}{2} \in (\alpha, 1)$. By definition we have:

$$\limsup_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \sup_{k\geq n} \sqrt[k]{|a_k|} = \alpha.$$

We know that $\limsup_{n\to\infty} \sqrt[n]{|a_n|} = \alpha$, so it holds that $\sup_{k\geq n} \sqrt[k]{|a_k|} \to \alpha$. Since this sequence is monotonically decreasing and converges to α, by definition we can find an N such that:

$$\sup_{k\geq N} \sqrt[k]{|a_k|} < q \implies |a_k| < q^k.$$

We can now compare,

$$\sum_{k=N}^{\infty} |a_k| < \sum_{k=N}^{\infty} q^k$$

2

We now apply the majorant test and know that the geometric series converges if $|q| < 1 \implies \alpha < 1$. Thus the series $\sum_{n=0}^{\infty} a_n$ converges absolutely for $\alpha < 1$.

Now let $\alpha > 1$. Since the limit superior is the largest accumulation point, there exists a subsequence $\sqrt[n_k]{|a_{n_k}|}$ that converges to $\alpha > 1$. In other words, the sequence $(a_n)_n$ cannot converge to zero, which is a necessary condition for convergence of the series. Therefore, the series does not converge and the theorem follows.

□

Example 3.1.14
Using the root test, investigate whether the series

$$\sum_{n=1}^{\infty} \frac{\log(2^n)}{e^n}$$

converges.

Proof. We rewrite

$$\sum_{n=1}^{\infty} \frac{\log(2^n)}{e^n} = \log(2) \sum_{n=1}^{\infty} \frac{n}{e^n}$$

and use the root test. We have

$$\lim_{n\to\infty} \sqrt[n]{\frac{n}{e^n}} = \lim_{n\to\infty} \frac{\sqrt[n]{n}}{\sqrt[n]{e^n}} = \frac{1}{e} \underbrace{\lim_{n\to\infty} \sqrt[n]{n}}_{\to 1} = \frac{1}{e} < 1.$$

Thus the series converges by the root test. □

Theorem 3.1.15 (D'Alembert Ratio Test)
Let (a_n) be a sequence of real numbers with $a_n \neq 0$ for all $n \in \mathbb{N}$, and assume that

$$\lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} = \alpha \text{ exists.}$$

Let $\alpha \geq 0$. Then

$$\alpha < 1 \implies \sum_{n=0}^{\infty} a_n \qquad \text{converges absolutely}$$

and

$$\alpha > 1 \implies \sum_{n=0}^{\infty} a_n \qquad \text{does not converge.}$$

Example 3.1.16
Using the ratio test, investigate whether the series

$$\sum_{n=1}^{\infty} \frac{n!}{n^n}$$

converges.

Solution: We use the ratio test. We have

$$\lim_{n\to\infty} \frac{\left|\frac{(n+1)!}{(n+1)^{n+1}}\right|}{\left|\frac{n!}{n^n}\right|} = \lim_{n\to\infty} \frac{(n+1)!}{n!} \frac{n^n}{(n+1)^{n+1}} = \lim_{n\to\infty} \frac{n+1}{n+1} \left(\frac{n}{n+1}\right)^n = \lim_{n\to\infty} \frac{1}{(1+1/n)^n} = \frac{1}{e}.$$

2

Since $1/e < 1$, the series converges absolutely by the ratio test.

Remark 3.1.17
Not all convergence tests are equally useful for every computation. There is a "sensible" order when applying these tests. This order is of course not always the best, but a good "rule of thumb".

1. Is (a_n) a null sequence at all?
2. Is a $(-1)^n$ present? $\rightsquigarrow$ Leibniz test
3. Is the quotient $\frac{|a_{n+1}|}{|a_n|}$ easily estimable (resp. factorials)? $\rightsquigarrow$ Ratio test
4. Are powers of n present? $\rightsquigarrow$ Root test
5. Majorant/ Minorant test
6. Condensation test

3.2 Series Calculations

Example 3.2.1
Calculate the value of the series

$$\sum_{n=1}^{\infty} \frac{(-1)^n 2^{n-1} + 1}{3^n}.$$

Solution: We rewrite so that we obtain two geometric series:

$$\sum_{n=1}^{\infty} \frac{(-1)^n 2^{n-1}+1}{3^n} = \sum_{n=1}^{\infty} \frac{(-1)^n 2^{n-1}}{3^n} + \sum_{n=1}^{\infty} \frac{1}{3^n} = \sum_{n=1}^{\infty} \frac{1}{2}\left(\frac{-2}{3}\right)^n + \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n.$$

For both summands we can calculate the values of the series using the known formula for the geometric series. However, we must note in each case that we start summing from $n = 1$ and therefore must subtract the term for $n = 0$.

$$\sum_{n=1}^{\infty} \frac{(-1)^n 2^{n-1}+1}{3^n} = \sum_{n=1}^{\infty} \frac{1}{2}\left(\frac{-2}{3}\right)^n + \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n = \frac{1}{2}\left(\frac{1}{1-(\frac{-2}{3})} - 1\right) + \left(\frac{1}{1-\frac{1}{3}} - 1\right) = \frac{3}{10}.$$

Example 3.2.2
Calculate the value of the series

$$\sum_{n=1}^{\infty} \frac{\log(2^n)}{2^n}.$$

Solution: We recognize

$$\sum_{n=1}^{\infty} \frac{\log(2^n)}{2^n} = \log(2) \sum_{n=1}^{\infty} \frac{n}{2^n} = \log(2) \sum_{n=1}^{\infty} n \left(\frac{1}{2}\right)^n.$$

We now derive a general formula for series of the form $S = \sum_{n=1}^{\infty} nq^n$. We obtain by considering

$$qS = \sum_{n=1}^{\infty} nq^{n+1} = \sum_{n=0}^{\infty} nq^{n+1} = \sum_{n=1}^{\infty} (n-1)q^n = \sum_{n=1}^{\infty} nq^n - \sum_{n=1}^{\infty} q^n = S - \left(\frac{1}{1-q} - 1\right) = S - \left(\frac{q}{1-q}\right).$$

Hence we have

$$S(q-1) = -\frac{1}{1-q} = \frac{q}{q-1} \iff S = \frac{q}{(q-1)^2} \quad \text{if } |q| < 1.$$

Thus for our series we have

$$\log(2) \sum_{n=1}^{\infty} n \left(\frac{1}{2}\right)^n = \log(2) \frac{\frac{1}{2}}{(\frac{1}{2}-1)^2} = 2\log(2).$$

3.2.1 Telescope Sums (including Partial Fraction Decomposition)

Telescoping sums are specific sums whose result depends only on the first and last term, because all terms in the middle cancel each other out. They are sums of the form

$$\sum_{i=0}^{n} (a_i - a_{i+1}) = (a_0 - a_1) + (a_1 - a_2) + \cdots + (a_{n-1} - a_n) + (a_n - a_{n+1}) = a_0 - a_{n+1}.$$

Then for the value of the series, we have

$$\sum_{i=0}^{\infty} a_i = \lim_{n\to\infty} \sum_{k=0}^{n} a_k = a_0 - \lim_{n\to\infty} a_{n+1}.$$

Such a telescoping series is therefore convergent exactly when the limit $\lim_{n\to\infty} a_{n+1}$ exists. To bring a series into a telescoping form, a partial fraction decomposition is often useful.

Partial Fraction Decomposition

If we have a fraction of polynomials, we want to decompose it into so-called partial fractions. For example, this looks like:

$$\frac{2x+3}{x^2-x-2} = \frac{1}{3}\left(\frac{7}{x-2} - \frac{1}{x+1}\right).$$

The general procedure is explained here once with an example:

- Factor the denominator polynomial into linear factors.

  In our example, we look for the linear factorization of $x^2 - x - 2$. We find the zeros $+2$ and -1. This yields the linear factorization

  $$x^2 - x - 2 = (x-2)(x+1)$$

- Set up the partial fraction ansatz.

  We now set up the partial fraction ansatz. For this we write:

  $$\frac{2x+3}{x^2-x-2} = \frac{A}{x-2} + \frac{B}{x+1}$$

  for constant coefficients $A, B \in \mathbb{R}$.

- Cross-multiply

2

  We now want to multiply by the denominator polynomial. Note that we need the linear factorization for this!

  $$2x+3 = \frac{A}{x-2}(x-2)(x+1) + \frac{B}{x+1}(x-2)(x+1) = A(x+1) + B(x-2)$$

- Compare coefficients

  We now want to compare the coefficients in front of the individual powers of x. For this, we fully expand the expression on the right side and bring the left side over.

  $$2x+3 = Ax + A + Bx - 2B = (A+B)x + A - 2B.$$

  Hence, by comparing coefficients, we have $2 = A + B$ and $3 = A - 2B$.

- Solve the linear system

  Now we just need to solve the linear system from the coefficient comparison.

  $$\begin{cases} 2 = A + B \\ 3 = A - 2B \end{cases} \qquad A = \frac{7}{3} \quad \text{and} \quad B = -\frac{1}{3}.$$

  Thus, for our partial fraction decomposition we have:

  $$\frac{2x+3}{x^2-x-2} = \frac{7/3}{x-2} + \frac{-1/3}{x+1} = \frac{1}{3}\left(\frac{7}{x-2} - \frac{1}{x+1}\right).$$

Example 3.2.3 (HS2021 M. Einsiedler)
Show that $\sum_{k=1}^{\infty} \frac{1}{4k^2-1} = \frac{1}{2}$.

Proof. We start with a partial fraction decomposition.

$$\sum_{k=1}^{\infty} \frac{1}{4k^2-1} = \frac{1}{2}\sum_{k=1}^{\infty}\left(\frac{1}{2k-1} - \frac{1}{2k+1}\right)$$

Now we recognize that this sum is a telescoping sum. All terms cancel except the first and the limit of the last. This can be proved by induction (1.1). Define for this $s_n = \sum_{k=1}^{n}\left(\frac{1}{2k-1} - \frac{1}{2k+1}\right)$. We want to show that $s_n = \frac{1}{2\cdot 1-1} - \frac{1}{2n+1}$. For this we use a base case

$$s_1 = \frac{1}{2\cdot 1-1} - \frac{1}{2\cdot 1+1}.$$

And an inductive step

$$s_{n+1} = \frac{1}{2\cdot 1-1} - \frac{1}{2n+1} + \frac{1}{2(n+1)-1} - \frac{1}{2(n+1)+1} = \frac{1}{2\cdot 1-1} - \frac{1}{2(n+1)+1}.$$

Thus we can compute:

$$\sum_{k=1}^{\infty}\left(\frac{1}{2k-1} - \frac{1}{2k+1}\right) = \lim_{n\to\infty}\left(\frac{1}{2\cdot 1-1} - \frac{1}{2n+1}\right) = \frac{1}{2\cdot 1-1} = 1.$$

The value of the series therefore evaluates to

$$\sum_{k=1}^{\infty} \frac{1}{4k^2-1} = \frac{1}{2}\underbrace{\sum_{k=1}^{\infty}\left(\frac{1}{2k-1} - \frac{1}{2k+1}\right)}_{=1} = \frac{1}{2}.$$

□

Example 3.2.4
Calculate the value of the following series using a telescoping sum

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}.$$

Solution: We start with a partial fraction decomposition. We have

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

The partial sums thus have the form

$$\sum_{n=1}^{k}\left(\frac{1}{n} - \frac{1}{n+1}\right) = 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3}\cdots - \frac{1}{k+1}.$$

Thus we see that the series is a telescoping series (this can again be proved by induction) and it follows for the limit

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \lim_{k\to\infty}\sum_{n=1}^{k}\left(\frac{1}{n} - \frac{1}{n+1}\right) = 1 - \lim_{k\to\infty}\frac{1}{k+1} = 1.$$

3.3 Power Series

Functions can be defined from convergent series.

$$f(x) = \sum_{k=0}^{\infty} a_k x^k$$

Such functions are called *power series*. For this definition to make sense, the series must converge absolutely. However, this convergence depends on which function value I substitute into my power series. The values x for which the series converges absolutely are called the *radius of convergence* of the power series.

3.3.1 Radius of Convergence

Remark 3.3.1 (What is the radius of convergence?)
We now have a series of the form

$$f(x) = \sum_{k=0}^{\infty} a_k x^k$$

and want to know for which x this series converges. For this we can use the root test as usual and find:

$$\limsup_{n\to\infty} \sqrt[n]{|a_n x^n|} = \limsup_{n\to\infty} \sqrt[n]{|a_n|}|x| < 1.$$

Thus we have the restriction on the x-values

$$|x| < \frac{1}{\limsup_{n\to\infty} \sqrt[n]{|a_n|}} := R$$

2

So in the end with this definition of R, we find that

$$-R < x < R.$$

Thus the series converges when x lies in the open interval $(-R, R)$. Exactly this is the radius of convergence of a power series!

Theorem 3.3.2 (Radius of Convergence)
Let $\sum_{k=0}^{\infty} a_k x^k$ be a power series. The **radius of convergence** of the series is the number $R \in \mathbb{R}_0^+$ or the symbol $R = \infty$, defined by

$$\rho = \limsup_{n\to\infty} \sqrt[n]{|a_n|} \qquad R = \begin{cases} 0 & \rho = \infty \\ \frac{1}{\rho} & 0 < \rho < \infty \\ \infty & \rho = 0 \end{cases}$$

Remark 3.3.3
The radius of convergence can also be calculated using the ratio test.

Example 3.3.4 (FS2023 G. Felder)
Calculate the radius of convergence R of the power series $\sum_{k=0}^{\infty} \left(1 + \frac{1}{n}\right)^{n^2} z^n$.

Solution: We calculate using the definition of the radius of convergence:

$$\rho = \limsup_{n\to\infty} \sqrt[n]{\left(1 + \frac{1}{n}\right)^{n^2}} = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^{n} = e.$$

Thus the radius of convergence is $R = \frac{1}{e}$.

What we wrote informally above, we now put into a theorem.

Theorem 3.3.5 (Convergence of Power Series)
Let $\sum_{k=0}^{\infty} a_k x^k$ be a power series with radius of convergence $R \in (0, \infty) \cup \{\infty\}$. The series $\sum_{k=0}^{\infty} a_k x^k$ converges absolutely for all $x \in \mathbb{R}$ with $|x| < R$, and does not converge for all $x \in \mathbb{R}$ with $|x| > R$.

In particular, for $x \in (-R, R)$ we can define the following function

$$f(x) = \sum_{k=0}^{\infty} a_k x^k.$$

Remark 3.3.6 (Power series are continuous)
Considering the sequence of functions $f_n(x) = \sum_{k=0}^{n} a_k x^k$, this sequence of functions converges *uniformly* to the power series within the radius of convergence. Since polynomials are continuous functions, power series always define *continuous functions* within their radius of convergence.

Multiple Choice Question 3.3.7 (FS2023, G. Felder)
Which of the following series $\sum_{k=1}^{\infty} a_k$ diverges to infinity?

(A)
$$\sum_{k=1}^{\infty} \frac{(-1)^k}{\log(k)}.$$

(B)
$$\sum_{k=1}^{\infty} (-1)^k.$$

(C)
$$\sum_{k=1}^{\infty} \frac{1}{\sqrt{k(k+1)}}.$$

(D)
$$\sum_{k=1}^{\infty} k^2 2^{-k}.$$

Solution:

(A) Converges by the Leibniz-Test.

(B) Alternates between -1 and 0.

(C) Diverges to infinity, by comparison with the harmonic series.

(D) Converges by the Cauchy root criterion.

Multiple Choice Question 3.3.8 (HS2022, G. Felder)
For each $n \in \mathbb{N}$ let $s_n = \sum_{j=1}^{n} \frac{1}{j}$. Which statement is correct?

(A) $(s_n)_n$ is monotonic and bounded hence convergent.

(B) $\left(\frac{1}{s_n}\right)_n$ converges to 0.

(C) The sequence $\left(\frac{1}{j}\right)_j$ is a null sequence, hence the series $\sum_{j=1}^{\infty} \frac{1}{j} = \lim_{n\to\infty} s_n$ converges.

2 (D) $(s_n)_n$ is a Cauchy sequence.

Solution:

(A) False. We know that the harmonic series diverges.

(B) True, because s_n diverges, $1/s_n$ converges to zero.

(C) False. We know that the harmonic series diverges. Being a Null sequence is only necessary, not sufficient.

(D) False. If it were Cauchy, it would converge and we know that the harmonic series diverges.

Chapter 4

Differentiable Functions

4.1 The Derivative Function

In the following, let $D \subseteq \mathbb{R}$ be a non-empty subset without isolated points, i.e., explicitly every $x \in D$ is an accumulation point of $D \setminus \{x\}$. Intuitively, an accumulation point of a set is a point where other points of the set can still be found arbitrarily close. A point $x \in D$ is called an accumulation point of D if in every neighborhood of x there is at least one point of D different from x.

Definition 4.1.1 (Derivative)
Let $f : D \to \mathbb{R}$ be a function and $x_0 \in D$. The function f is **differentiable** at the point x_0 if the limit

$$f'(x_0) := \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

exists. The value $f'(x_0)$ is the **derivative of the function f at the point x_0**. If f is differentiable at every point $x \in D$, then f is **differentiable on D** and the function $f' : D \to \mathbb{R}$ is the **derivative** of f.

There are many different interpretations of the derivative of a function. Here are some of the most important ones:

- The derivative at a point x_0 is the value of the slope of a tangent line to the graph of f at that point.
- The derivative function is the best local linear approximation of the function f.

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + o(|x - x_0|)$$

Example 4.1.2 (Difference Quotient)
Using the definition of the derivative, show that the function $f : \mathbb{R} \to \mathbb{R}$ defined as

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

is differentiable at the origin.
Solution: We form the difference quotient at 0

$$\lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0} \frac{x^2 \sin\left(\frac{1}{x}\right)}{x} = \lim_{x \to 0} x \sin\left(\frac{1}{x}\right) = \lim_{x \to \pm\infty} \frac{\sin(x)}{x} = 0.$$

Thus the derivative $f'(0) = 0$ is well-defined and f is therefore differentiable at the origin.

Example 4.1.3 (Continuous $\not\Longrightarrow$ Differentiable)
Show that the function $f : \mathbb{R} \mapsto \mathbb{R}$ defined by

$$f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

is continuous but not differentiable at the origin.
Solution: To briefly show continuity, we need to compute the limit towards zero.

$$\lim_{x \to 0} x \sin\left(\frac{1}{x}\right) = \lim_{x \to \pm\infty} \frac{\sin(x)}{x} = 0.$$

Because the right-hand and left-hand limits coincide with the function value, f is continuous at 0.

Now we compute the difference quotient:

$$\lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0} \frac{x \sin\left(\frac{1}{x}\right)}{x} = \lim_{x \to 0} \sin\left(\frac{1}{x}\right) = \lim_{x \to \pm\infty} \sin(x).$$

This limit is not well-defined. Therefore the derivative at the origin does not exist and hence f is not differentiable there.

2 **Remark 4.1.4** (Differentiability $\Longrightarrow$ Continuity)
However, the converse holds, i.e., differentiability at x_0 always implies continuity at that point. Because we can verify sequential continuity:

$$\lim_{x \to x_0} f(x) = \lim_{x \to x_0} \Big(f(x) - f(x_0) + f(x_0) \Big) = \lim_{x \to x_0} \left(\underbrace{\frac{f(x) - f(x_0)}{x - x_0}}_{\to f'(x_0) < \infty} \cdot \underbrace{(x - x_0)}_{\to 0} + f(x_0) \right) = f(x_0)$$

4.1.1 Differentiation Rules

Sum and Product Rule

Theorem 4.1.5 (Sum and Product Rule)
Let $D \subseteq \mathbb{R}$ and $x_0 \in D$ be an accumulation point of $D \setminus \{x_0\}$. Let $f, g : D \to \mathbb{R}$ be differentiable at the point x_0. Then $f + g$ and $f \cdot g$ are differentiable at x_0 and we have:

$$(f + g)'(x_0) = f'(x_0) + g'(x_0),$$
$$(f \cdot g)'(x_0) = f'(x_0)g(x_0) + f(x_0)g'(x_0).$$

In particular, any multiple of f is differentiable at x_0 and $(\alpha f)'(x_0) = \alpha f'(x_0)$ for all $\alpha \in \mathbb{R}$.

Chain Rule

Theorem 4.1.6 (Chain Rule (Exam))
Let $D, E \subseteq \mathbb{R}$ be subsets and $x_0 \in D$ an accumulation point of $D\backslash\{x_0\}$. Let $f : D \to E$ be differentiable at x_0, such that $y_0 = f(x_0)$ is an accumulation point of $E\backslash\{y_0\}$ and $g : E \to \mathbb{R}$ is differentiable at y_0. Then $g \circ f : D \to \mathbb{R}$ is differentiable at x_0 and we have

$$(g \circ f)'(x_0) = g'(f(x_0)) \cdot f'(x_0)$$

Proof. We can write the function $g(y)$ as:

$$g(y) = g(y_0) + g'(y_0)(y - y_0) + \omega(y)(y - y_0)$$

$$\omega(y) = \begin{cases} \frac{g(y)-g(y_0)}{y-y_0} - g'(y_0) & \text{if } y \in E \setminus \{x_0\} \\ 0 & \text{if } y = y_0 \end{cases}$$

Because the function g is differentiable at y_0, ω is continuous. Consider the limit $y \to y_0$, the left term then becomes the difference quotient. The differentiability of g ensures that the limit exists.

Set $y = f(x)$ and note that $y_0 = f(x_0)$:

$$g(f(x)) = g(f(x_0)) + g'(f(x_0))[f(x) - f(x_0)] + \omega(f(x))[f(x) - f(x_0)]$$

From this equation we „build" the difference quotient. We subtract $g(f(x_0))$ and divide by $x - x_0$

$$\frac{g(f(x)) - g(f(x_0))}{x - x_0} = g'(f(x_0))\frac{f(x) - f(x_0)}{x - x_0} + \omega(f(x))\frac{f(x) - f(x_0)}{x - x_0}$$

2

More precisely, for $\omega(f(x)$ due to the differentiability of g at y_0:

$$\lim_{x \to x_0} \omega(f(x)) = \lim_{x \to x_0} \left(\underbrace{\frac{g(f(x)) - g(f(x_0))}{f(x) - f(x_0)}}_{\to g'(f(x_0))} - g'(f(x_0)) \right) = 0$$

Therefore, for the limit as $x \to x_0$:

$$\lim_{x \to x_0} \left(\frac{g(f(x)) - g(f(x_0))}{x - x_0} \right) = \lim_{x \to x_0} \left(g'(f(x_0)) \underbrace{\frac{f(x) - f(x_0)}{x - x_0}}_{\to f'(x_0)} + \underbrace{\omega(f(x))}_{\to 0} \frac{f(x) - f(x_0)}{x - x_0} \right)$$
$$= g'(f(x_0))f'(x_0)$$

□

Quotient Rule

Theorem 4.1.7 (Quotient Rule)
Let $D \subseteq \mathbb{R}$ be a subset and $x \in D$ an accumulation point of $D \setminus \{x_0\}$ and let $f, g : D \to \mathbb{R}$ be differentiable at x_0. If $g(x_0) \neq 0$, then f/g is differentiable at x_0 and we have

$$\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g^2(x_0)}.$$

Derivative of the Inverse

Theorem 4.1.8 (Derivative of the Inverse (Exam))
Let $D, E \subseteq \mathbb{R}$ and let $f : D \to E$ be a continuous, bijective function whose inverse $f^{-1} : E \to D$ is also continuous. Let $\bar{x} \in D$ be an accumulation point of $D \setminus \{\bar{x}\}$ and let f be differentiable at $\bar{x}$ with $f'(\bar{x}) \neq 0$. Then f^{-1} is differentiable at $\bar{y} := f(\bar{x})$ and we have

$$(f^{-1})'(\bar{y}) = \frac{1}{f'(\bar{x})}.$$

Proof. To compute $(f^{-1})'(\bar{y})$, take an arbitrary sequence $(y_n)_{n=0}^{\infty} \subset E \setminus \{\bar{y}\}$ such that $y_n \to \bar{y}$ and set $x_n = f^{-1}(y_n)$. Then

$$\frac{f^{-1}(y_n) - f^{-1}(\bar{y})}{y_n - \bar{y}} = \frac{x_n - \bar{x}}{f(x_n) - f(\bar{x})} = \left(\frac{f(x_n) - f(\bar{x})}{x_n - \bar{x}}\right)^{-1}.$$

By the continuity of f^{-1}, we have $x_n \to \bar{x} = f^{-1}(\bar{y})$. Because f is differentiable at $\bar{x}$ and $f'(\bar{x}) \neq 0$, the difference quotient for f converges to $f'(\bar{x})$. Therefore, the reciprocal of this difference quotient converges to the reciprocal of $f'(\bar{x})$. That is,

$$\left(\frac{f(x_n) - f(\bar{x})}{x_n - \bar{x}}\right)^{-1} \longrightarrow \frac{1}{f'(\bar{x})}.$$

Consequently,

$$\lim_{n\to\infty} \frac{f^{-1}(y_n) - f^{-1}(\bar{y})}{y_n - \bar{y}} = \frac{1}{f'(\bar{x})}.$$

2

Since the sequence $(y_n)_{n=0}^{\infty}$ was chosen arbitrarily, this implies that the limit of the difference quotient for f^{-1} exists as y approaches $\bar{y}$. Hence,

$$\lim_{y\to\bar{y}} \frac{f^{-1}(y) - f^{-1}(\bar{y})}{y - \bar{y}} = \frac{1}{f'(\bar{x})},$$

which is exactly $(f^{-1})'(\bar{y})$. □

Remark 4.1.9
It is important to note that f' and $(f^{-1})'$ are evaluated at different points, i.e., $\bar{y} = f(\bar{x})$.

4.2 Important Theorems of Differential Calculus

4.2.1 Local Extrema

Definition 4.2.1 (Local Extrema)
Let $D \subseteq \mathbb{R}$ and $x_0 \in D$. We say that a function $f : D \to \mathbb{R}$ has a **local maximum** at x_0 if there exists $\delta > 0$ such that $f(x) \leq f(x_0)$ for all $x \in D \cap (x_0 - \delta, x_0 + \delta)$. If the inequality is strict, i.e., $f(x) < f(x_0)$ for all $x \in D \cap (x_0 - \delta, x_0 + \delta) \setminus \{x_0\}$, then f has an **isolated** local maximum at x_0. The value $f(x_0)$ is called a **local maximum value** of f.

A **local minimum**, an **isolated** local minimum and a **local minimum value** of f are defined analogously. Furthermore, we call x_0 a *local extremum* of f and $f(x_0)$ a **local extremal value** of f if f has a local minimum or a local maximum at x_0.

Theorem 4.2.2 (Derivative and Extrema)
Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$. Suppose f has a local extremum at $x_0 \in D$, is differentiable, and x_0 is both a right-hand and a left-hand accumulation point of D (explicitly, x_0 is an accumulation point for $D \cap (x_0, \infty)$ and $D \cap (-\infty, x_0)$). Then $f'(x_0) = 0$.

Thus we know that for a function $f : I \to \mathbb{R}$ on an interval I, for a local extremum $x_0 \in I$ one of the following holds:

1. $x_0 \in I$ is a boundary point of I
2. f is not differentiable at x_0
3. f is differentiable at x_0 and $f'(x_0) = 0$

More precisely, all local extrema of a differentiable function on an open interval are zeros of the derivative.

4.2.2 Rolle's Theorem

Theorem 4.2.3 (Rolle's Theorem)
Let $a < b$ and $f : [a, b] \to \mathbb{R}$ continuous and differentiable on the open interval (a, b). If $f(a) = f(b)$, then there exists a $\xi \in (a, b)$ such that $f'(\xi) = 0$.

4.2.3 Mean Value Theorem

Theorem 4.2.4 (Mean Value Theorem)
Let $a < b$ and $f : [a, b] \to \mathbb{R}$ continuous and differentiable on the open interval (a, b). Then there exists a $\xi \in (a, b)$ such that:

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}$$

4.3 Bernoulli-de L'Hospital's Rule

Theorem 4.3.1
Let $I \subseteq \mathbb{R}$ be an open interval and $a \in I \cup \{\pm\infty\}$. Let $f, g : I \setminus \{a\} \to \mathbb{R}$ be differentiable with $g'(x) \neq 0$ for all $x \in I \setminus \{a\}$. Assume one of the following conditions holds:

$$\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = 0 \quad \lim_{x \to a} |f(x)| = \lim_{x \to a} |g(x)| = +\infty.$$

If the limit

$$\lim_{x \to a} \frac{f'(x)}{g'(x)} = L \quad (L \in \mathbb{R} \cup \{\pm\infty\}),$$

exists, then:

$$\lim_{x \to a} \frac{f(x)}{g(x)} = L.$$

Example 4.3.2
Compute the limit

$$\lim_{x \to 0} (e^{\alpha x} + x)^{\frac{1}{x}}$$

for all $\alpha \in \mathbb{R}$.
Solution: Since exp() and log() are continuous functions we write

$$\lim_{x\to 0} (e^{\alpha x} + x)^{\frac{1}{x}} = \exp\left(\lim_{x\to 0} \frac{1}{x} \log\left(e^{\alpha x} + x\right)\right).$$

Now we compute the inner limit using Bernoulli-de L'Hospital's rule.

$$\lim_{x\to 0} \frac{\log\left(e^{\alpha x} + x\right)}{x} = \lim_{x\to 0} \frac{\alpha e^{\alpha x} + 1}{e^{\alpha x} + x} = \alpha + 1.$$

Thus for the limit we have

2

$$\lim_{x\to 0} (e^{\alpha x} + x)^{\frac{1}{x}} = e^{\alpha+1}.$$

Example 4.3.3 (FS2024, A. Figalli)
Compute the limit.

$$\lim_{x\to 0} \frac{x + \cos(x)\sin(3x)}{\sin^3(x) + 3\sin(x) - x}.$$

Solution: We use Bernoulli-de L'Hospital's rule and find

$$\lim_{x\to 0} \frac{x + \cos(x)\sin(3x)}{\sin^3(x) + 3\sin(x) - x} = \lim_{x\to 0} \frac{1 - \sin(x)\sin(3x) + 3\cos(x)\cos(3x)}{3\cos(x)\sin^2(x) + 3\cos(x) - 1} = \frac{1+3}{3-1} = 2.$$

4.4 Convex Functions

Definition 4.4.1 (Convex Function)
Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \to \mathbb{R}$. Then we call f *convex* if for all $a, b \in I$ with $a < b$ and all $t \in (0, 1)$, the inequality

$$f\big((1-t)a + tb\big) \leq (1-t)f(a) + tf(b)$$

holds. We say that f is *strictly convex* if the inequality is strict.

A function $g : I \to \mathbb{R}$ is called (strictly) *concave* if $-g$ is (strictly) convex.

Remark 4.4.2
3 There are other equivalent convexity criteria to the definition given above. As an example, here is another one:

$$\frac{f(x) - f(a)}{x - a} \leq \frac{f(b) - f(x)}{b - x} \quad \forall x \in (a, b) \subseteq I.$$

Theorem 4.4.3 (Convexity and the First Derivative (Exam))
Let I be an interval and $f : I \to \mathbb{R}$ a differentiable function. Then f is convex if and only if f' is monotonically increasing.

Proof.

$\Longleftarrow$ First assume that f' is monotonically increasing and show that f must be convex. Let $a < b$ and $x \in (a, b)$. By the Mean Value Theorem, we know there exists a $\xi \in (a, x)$ and

a $\zeta \in (x, b)$ such that

$$f'(\xi) = \frac{f(x) - f(a)}{x - a}, \qquad f'(\zeta) = \frac{f(b) - f(x)}{b - x}$$

Because f' is monotonically increasing, we have: $\xi \leq \zeta \implies \frac{f(x)-f(a)}{x-a} \leq \frac{f(b)-f(x)}{b-x}$. Hence f is convex.

$\implies$ Comparing the two inequalities we obtain:

$$\frac{f(a+h) - f(a)}{h} \leq \frac{f(b) - f(b-h)}{h}$$

Considering the limit as $h \to 0^+$, both terms become derivatives. For the right term it is not directly obvious. It follows via a simple transformation with $\tilde{h} = -h$:

$$\frac{f(b) - f(b-h)}{h} = \frac{f(b) - f(b+\tilde{h})}{-\tilde{h}} = \frac{f(b+\tilde{h}) - f(b)}{\tilde{h}}.$$

Thus in the limit $h \to 0^+$ we get: $f'(a) \leq f'(b)$. Since $a < b$ were chosen arbitrarily, the theorem follows.

□

Remark 4.4.4
If f is strictly convex, then f' is also strictly monotonically increasing. However, the converse of this statement does not hold! Consider for example x^3.

3

Multiple Choice Question 4.4.5 (FS2023, G. Felder)
The function $f : x \mapsto \max(x, 1 - x)$ is

(A) not continuous.

(B) continuous, but not differentiable.

(C) differentiable, but not continuously differentiable.

(D) continuously differentiable, but not twice differentiable.

Solution: The function is constructed as the upper part of the intersection of the two lines x and $1 - x$. This automatically implies that (B) is correct.

Multiple Choice Question 4.4.6 (HS2023, A. Figalli)
Let $f : [0, \infty) \to [0, \infty)$ be a continuous function such that $f(0) = 0$. Which of the following implications is always correct?

(A) If f is uniformly continuous, then f is bounded.

(B) If f is unbounded, then f is surjective.

(C) If f is monotonic, then f is injective.

(D) If f is Lipschitz continuous, then f is differentiable.

Solution:

(A) False. Consider $f(x) = x$.

(B) True, because $f(0) = 0$.

(C) False. Consider $f(x) = 0$.

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(D) False. Consider any non-differentiable, bounded, continuous function. e.g., similar to the function in the previous question.

Chapter 5

The Riemann-Integral

5.1 Partitions and Step Functions

Definition 5.1.1 (Riemann Partition)
A **Riemann partition** of $[a, b]$ is a finite set of points

$$a = x_0 < x_1 < \cdots < x_n = b$$

with $n \in \mathbb{N}$. The points $x_0, \ldots, x_n$ are the partition points.

Definition 5.1.2 (Step Function)
A function $f : [a, b] \to \mathbb{R}$ is a **step function** if there exists a partition $a = x_0 < \cdots < x_n = b$ of $[a, b]$ such that for $k = 1, 2, \ldots, n$ the restriction of f to the open interval (x_{k-1}, x_k) is constant.

5.2 Definition of the Riemann Integral

Definition 5.2.1 (Riemann Integral of a Step Function)
Let $f : [a, b] \to \mathbb{R}$ be a step function with respect to the partition $a = x_0 < \cdots < x_n = b$ of $[a, b]$. Then the integral of f on $[a, b]$ is defined as the real number

$$\int_a^b f(x)dx = \sum_{k=1}^{n} c_k(x_k - x_{k-1}),$$

where c_k corresponds to the function value of f on the interval (x_k, x_{k-1}).

The integral of step functions does **not** depend on the partition of the interval. Furthermore, it is **linear** and **monotone**.

Definition 5.2.2 (Lower and Upper Sums)
Let $f : [a, b] \to \mathbb{R}$ be a function. Then the set of **lower sums** $\mathcal{L}(f) \subseteq \mathbb{R}$ and **upper sums** $\mathcal{U}(f) \subseteq \mathbb{R}$ of f is

$$\mathcal{L}(f) = \left\{ \int_a^b \ell \, dx \;\middle|\; \ell \text{ a step function and } \ell \leq f \right\}$$

$$\mathcal{U}(f) = \left\{ \int_a^b u \, dx \;\middle|\; u \text{ a step function and } u \geq f \right\}.$$

Definition 5.2.3 (The Riemann Integral)
A bounded function $f : [a, b] \to \mathbb{R}$ is called *Riemann-integrable* if $\sup \mathcal{L}(f) = \inf \mathcal{U}(f)$. In this case, this common value is the Riemann integral of f and is expressed by

$$\int_a^b f(x) \, dx = \sup \mathcal{L}(f) = \inf \mathcal{U}(f).$$

Remark 5.2.4
The Riemann integral is also **linear** and **monotone**. It also satisfies a triangle inequality:

$$\left| \int_a^b f(x) \, dx \right| \leq \int_a^b |f(x)| \, dx.$$

This inequality is central to many applications and estimates!

3

5.3 Integrability Conditions

Theorem 5.3.1 (Monotone Functions are Integrable)
Every monotonic function $f : [a, b] \to \mathbb{R}$ is Riemann-integrable.

Additional Remark: Every bounded and piecewise monotonic function on $[a, b]$ is also Riemann-integrable.

Theorem 5.3.2 (Continuous Functions are Integrable (Exam))
Every continuous function $f : [a, b] \to \mathbb{R}$ is Riemann-integrable.

Proof. Since the interval $[a, b]$ is compact, we know that f is uniformly continuous. Thus for every $\varepsilon > 0$ there exists a uniform $\delta > 0$ such that:

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$$

for all $x, y \in [a, b]$.

Now we choose a partition of $[a, b]$ as

$$a = x_0 < x_1 < \ldots < x_{n-1} < x_n = b \qquad x_k - x_{k-1} < \delta.$$

We now define for $k \in \{1, \ldots, n\}$:

$$c_k = \min\{f(x) \mid x_{k-1} \leq x \leq x_k\} \qquad d_k = \max\{f(x) \mid x_{k-1} \leq x \leq x_k\}.$$

The maxima and minima are well-defined because each interval is compact and f is continuous.

Now define the lower and upper step functions for $x \in [a, b]$ as:

$$\ell(x) = \begin{cases} c_k & \text{if } x \in [x_{k-1}, x_k) \\ c_n & \text{if } x = b \end{cases} \qquad u(x) = \begin{cases} d_k & \text{if } x \in [x_{k-1}, x_k) \\ d_n & \text{if } x = b \end{cases}$$

We have $\ell \leq f \leq u$ and due to the uniform continuity of f, we have $d_k - c_k < \varepsilon$, because we chose $x_k - x_{k-1} < \delta$. Therefore:

$$\int_a^b (u - \ell)(x)dx = \sum_{k=1}^{n} \underbrace{(d_k - c_k)}_{<\varepsilon}(x_k - x_{k-1}) < \varepsilon \sum_{k=1}^{n}(x_k - x_{k-1}) = \varepsilon(b - a)$$

and the theorem follows, since $\varepsilon > 0$ was chosen arbitrarily. □

Additional Remark: Every piecewise continuous function on $[a, b]$ is also Riemann-integrable.

5.4 Theorems on Riemann Integrability

Theorem 5.4.1 (Uniform Convergence and the Riemann Integral (Exam))
Let (f_n) with $f_n : [a, b] \to \mathbb{R}$ be a sequence of integrable functions that converges uniformly to a function $f : [a, b] \to \mathbb{R}$. Then f is integrable and

$$\int_a^b f(x)\, dx = \lim_{n\to\infty} \int_a^b f_n(x)\, dx.$$

3 *Proof.* Fix $\varepsilon > 0$. By uniform convergence, there exists N such that $|f_n - f| \leq \varepsilon$ on $[a, b]$ for all $n \geq N$.
Since f_N is integrable, there exist step functions ℓ, u with $\ell \leq f_N \leq u$ and $\int_a^b (u - \ell)\, dx < \varepsilon$. Set $\hat{\ell} = \ell - \varepsilon$ and $\hat{u} = u + \varepsilon$. Then both $\hat{\ell}$ and $\hat{u}$ are step functions. Also, since $|f_N - f| < \varepsilon$,

$$\hat{\ell} = \ell - \varepsilon \leq f_N - \varepsilon \leq f \leq f_N + \varepsilon \leq u + \varepsilon = \hat{u}$$

and (because $\hat{u} - \hat{\ell} = u - \ell + 2\varepsilon$)

$$\int_a^b (\hat{u} - \hat{\ell})\, dx = \int_a^b (u - \ell)\, dx + 2\varepsilon(b - a) < \varepsilon + 2\varepsilon(b - a).$$

As $\varepsilon > 0$ is arbitrary, we find that f is integrable.
Moreover, using monotonicity of the Riemann Integral and the triangle inequality we find,

$$\left| \int_a^b f\, dx - \int_a^b f_n\, dx \right| = \left| \int_a^b (f - f_n)\, dx \right| \leq \int_a^b |f - f_n|\, dx \leq \varepsilon(b - a) \quad \forall n \geq N.$$

□

Remark 5.4.2
In other words: The Riemann integral and the limit **commute** under uniform convergence.

Remark 5.4.3 (Mean Value Theorem for Integrals)
There also exists a mean value theorem for integrals, which was introduced in the problem sets. Accordingly, for a continuous function f on a compact interval $[a, b]$, there always exists a value $\xi \in [a, b]$ such that

$$\int_a^b f(x)dx = f(\xi)(b - a).$$

Geometrically interpreted, there always exists a function value $f(\xi)$ such that the resulting rectangle over $[a, b]$ has the same area as the Riemann integral of f over $[a, b]$.

Proof. Since f is continuous, f attains a maximum $M = f(x_M)$ and a minimum $m = f(x_m)$ on $[a, b]$. We can therefore estimate

$$m(b-a) \leq \int_a^b f(x)dx \leq M(b-a) \iff f(x_m) \leq \frac{1}{b-a}\int_a^b f(x)dx \leq f(x_M).$$

Because f is continuous, by the Intermediate Value Theorem there exists a ξ between x_m and x_M with

$$f(\xi) = \frac{1}{b-a}\int_a^b f(x)dx.$$

□

Multiple Choice Question 5.4.4 (FS2025, L. Kobel-Keller)
Given a function $f : [a, b] \to \mathbb{R}$ with $-\infty < a < b < \infty$. Which of the following statements are correct?

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(A) The function f is Riemann-integrable over $[a, b]$ if it is differentiable on the whole interval $[a, b]$.

(B) For the function f to be Riemann-integrable over $[a, b]$, it must not have any points of discontinuity on $[a, b]$.

(C) If the function f is Riemann-integrable over $[a, b]$, then the function is also differentiable on $[a, b]$.

(D) The considered function is Riemann-integrable over $[a, b]$ as long as it attains only finitely many different function values.

Solution:

(A) True. f is in particular continuous, and every continuous function is Riemann-integrable on a compact interval.

(B) False. Consider a step function.

(C) False. Consider a step function.

(D) False. Consider for example the Dirichlet function $f(x) = \chi_{\mathbb{Q}}(x)$.

Chapter 6

Differential- and Integral Calculus

6.1 Fundamental Theorem of Calculus

Definition 6.1.1 (Antiderivative)
Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ a function. Any differentiable function $F : I \to \mathbb{R}$ with the property that $F' = f$ is called an **antiderivative** of f.

Remark 6.1.2
An antiderivative is generally not unique. There also exist functions that have no antiderivative. There are also functions that are Riemann-integrable but have no antiderivative. An example is the function $f(x) = \operatorname{sgn}(x)$ – it has no antiderivative, but is Riemann-integrable on every bounded interval.

Theorem 6.1.3 (Fundamental Theorem of Calculus (Exam))
Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Then for every constant $C \in \mathbb{R}$ the function $F : [a, b] \to \mathbb{R}$ is an antiderivative of f. Furthermore, every antiderivative has the following form with a constant $C \in \mathbb{R}$:

$$F(x) = \int_a^x f(t)dt + C$$

3

Proof. Since f is continuous, we know that f is integrable on $[a, b]$. We first want to show that $F(x)$ is an antiderivative of f. We know, due to the continuity of f, that

$$|x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon$$

Now consider the δ-interval $(x_0, x_0 + \delta) \cap [a, b]$.

$$\begin{aligned}
\left|\frac{F(x) - F(x_0)}{x - x_0} - f(x_0)\right| &= \left|\frac{1}{x - x_0}\left(\int_a^x f(t)dt - \int_a^{x_0} f(t)dt\right) - f(x_0)\right| \\
&= \left|\frac{1}{x - x_0}\left(\int_{x_0}^x f(t)dt\right) - \frac{1}{x - x_0}\int_{x_0}^x f(x_0)dt\right| \\
&= \left|\frac{1}{x - x_0}\left(\int_{x_0}^x f(t) - f(x_0)dt\right)\right| \\
&\leq \frac{1}{x - x_0}\int_{x_0}^x \underbrace{|f(t) - f(x_0)|}_{<\varepsilon} dt \\
&< \varepsilon \, \frac{1}{x - x_0}\int_{x_0}^x dt = \varepsilon
\end{aligned}$$

Next, the δ-interval $(x_0 - \delta, x_0) \cap [a, b]$. The estimate proceeds completely analogously.

$$\begin{aligned}
\left|\frac{F(x) - F(x_0)}{x - x_0} - f(x_0)\right| &= \left|\frac{1}{x - x_0}\left(\int_a^x f(t)dt - \int_a^{x_0} f(t)dt\right) - f(x_0)\right| \\
&= \left|\frac{1}{x - x_0}\left(-\int_x^{x_0} f(t)dt\right) - \frac{1}{x_0 - x}\int_x^{x_0} f(x_0)dt\right| \\
&= \left|\frac{1}{x_0 - x}\left(\int_x^{x_0} f(t) - f(x_0)dt\right)\right| \\
&\leq \frac{1}{x_0 - x}\int_x^{x_0} \underbrace{|f(t) - f(x_0)|}_{<\varepsilon} dt \\
&< \varepsilon\,\frac{1}{x_0 - x}\int_x^{x_0} dt = \varepsilon
\end{aligned}$$

Since $\varepsilon > 0$ was chosen arbitrarily, we have: $\lim_{x \to x_0} \frac{F(x) - F(x_0)}{x - x_0} = F'(x_0) = f(x_0)$. Thus $F(x_0)$ is an antiderivative of f. The uniqueness up to the constant C follows:

$$\left(F(x) - \int_a^x f(t)dt\right)' = f(x) - f(x) = 0,$$

which means the only difference lies in a constant. The theorem follows. □

3

Remark 6.1.4
Through the Fundamental Theorem, we can now Riemann-integrate functions much more easily. For definite integrals, it follows directly from the Fundamental Theorem that

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

6.2 Integration Methods

6.2.1 Integration by Parts

Theorem 6.2.1 (Integration by Parts)
Let $f, g : [a, b] \to \mathbb{R}$ be continuously differentiable functions. Then

$$\int_a^b f(x)g'(x)\,dx = [f(x)g(x)]_a^b - \int_a^b f'(x)g(x)\,dx.$$

Example 6.2.2
Using this theorem, one can for example integrate $\log(x)$.

$$\int \log(x)\,dx = \int \log(x)(x)'\,dx = x\log(x) - \int x \cdot \frac{1}{x}\,dx = x\log(x) - x + C.$$

6.2.2 Integration by Substitution

Theorem 6.2.3 (Integration by Substitution (1st form))
Let $I, J \subseteq \mathbb{R}$ be intervals, $f : I \to J$ continuously differentiable and $g : J \to \mathbb{R}$ continuous. Then for all $[a, b] \subseteq I$,

$$\int_a^b g(f(x))f'(x)\,dx = \int_{f(a)}^{f(b)} g(y)\,dy.$$

Theorem 6.2.4 (Integration by Substitution (2nd form))
Let $I, J \subseteq \mathbb{R}$ be intervals, $f : I \to J$ continuously differentiable and $g : J \to \mathbb{R}$ continuous. Let $[a, b] \subseteq I$; assume that $f'(x) \neq 0$ for all $x \in [a, b]$, and let $f^{-1} : [f(a), f(b)] \to \mathbb{R}$ be the inverse function of $f\big|_{[a,b]}$. Then

$$\int_a^b g(f(x))\,dx = \int_{f(a)}^{f(b)} g(y)(f^{-1})'(y)\,dy.$$

Example 6.2.5 (Substitution)
We seek the indefinite integral

$$\int \frac{x}{1+x^2}\,dx.$$

3

We substitute $u = x^2$, then we have

$$\frac{du}{dx} = 2x \quad \Longleftrightarrow \quad du = 2x\,dx$$

and the integral becomes

$$\int \frac{x}{1+x^2}\,dx = \int \frac{x}{1+x^2}\frac{1}{2x}du = \frac{1}{2}\int \frac{1}{1+u}\,du = \frac{1}{2}\log|1+u| + C = \frac{1}{2}\log|1+x^2| + C.$$

Example 6.2.6 (Substitution)
We seek the indefinite integral

$$\int x\cos(1+x^2)\,dx.$$

We substitute $u = 1 + x^2$. Then we have

$$\frac{du}{dx} = 2x \quad \Longleftrightarrow \quad du = 2x\,dx$$

and for the integral:

$$\int x\cos(1+x^2)\,dx = \frac{1}{2}\int \cos(u)\,du = \frac{1}{2}\sin(u) + C = \frac{1}{2}\sin(1+x^2) + C.$$

6.3 Integral Test for Series

Theorem 6.3.1 (Integral Test for Series)
Let $f : [0, \infty) \to \mathbb{R}_{\geq 0}$ be a monotonically decreasing function. Then

$$\sum_{n=1}^{N+1} f(n) \leq \int_0^{N+1} f(x)\, dx \leq \sum_{n=0}^{N} f(n).$$

More precisely,

$$\sum_{n=1}^{\infty} f(n) \leq \int_0^{\infty} f(x)\, dx \leq \sum_{n=0}^{\infty} f(n),$$

so the series $\sum_{n=1}^{\infty} f(n)$ converges if the integral $\int_0^{\infty} f(x)\, dx$ converges.

Example 6.3.2 (Divergence of the Harmonic Series)
Thus we can easily show that the harmonic series diverges. Defining the function $f(x) = \frac{1}{1+x}$, it follows from the integral test that

$$\sum_{n=0}^{\infty} \frac{1}{n+1} = \sum_{n=1}^{\infty} \frac{1}{n} \geq \int_0^{\infty} \frac{1}{x+1}\, dx = \lim_{b \to \infty} \log(b+1) = +\infty.$$

Example 6.3.3 (Convergence of the Basel Problem)
Similarly, we can show that the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

3

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \sum_{n=2}^{\infty} \frac{1}{n^2} = 1 + \sum_{n=1}^{\infty} \frac{1}{(n+1)^2} \leq 1 + \int_0^{\infty} \frac{1}{(x+1)^2}\, dx = 1 - \lim_{b \to \infty} \left(\frac{1}{b+1} - 1 \right) = 2.$$

The exact series value is $\frac{\pi^2}{6} = 1.64493\ 40668\ 48226 \ldots \leq 2$.

6.4 Integration and Differentiation of Power Series

Theorem 6.4.1 (Integration and Differentiation of Power Series)
Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence $R > 0$. Then the power series

$$F(x) = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$$

also has radius of convergence R and is an antiderivative of f on $(-R, R)$.

From this, the analogous statement for the derivative of power series follows directly as a corollary.

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^n \qquad \forall x \in (-R, R).$$

Example 6.4.2 (Series Calculation by Differentiation)
Show that for $|x| < 1$ we have

$$\sum_{n=0}^{\infty} n x^n = \frac{x}{(1-x)^2}.$$

Proof. We know that the value of the geometric series is given by

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

Now we differentiate this power series and obtain

$$\sum_{n=0}^{\infty} nx^{n-1} = \frac{1}{(1-x)^2}.$$

We transform the series into the desired form by multiplying by x, and obtain

$$\sum_{n=0}^{\infty} nx^n = \frac{x}{(1-x)^2}.$$

□

6.5 Taylor Series and Analytic Functions

The derivative $f'(x_0)$ of a differentiable function f provides the possibility to approximate the function f locally in a small neighborhood of x_0 linearly. Here, the error term is $o(|x - x_0|)$ as $x \to x_0$.

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + o(|x - x_0|)$$

This local approximation of f can be "improved" – i.e., the error term reduced – by using polynomials of higher degree. This leads to the theory of Taylor expansions.

3 We call the polynomial

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$$

the *Taylor polynomial of degree n*. Here, the Taylor polynomial of degree 1 corresponds exactly to the linear approximation above.

$$P_1(x) = f(x_0) + f'(x_0)(x - x_0)$$

6.5.1 Taylor Expansion and Remainder Terms

Theorem 6.5.1 (Taylor Expansion of Order n with Integral Remainder Term)
Let $n \geq 1$, $f : [a, b] \to \mathbb{R}$ an n-times continuously differentiable function and let $x_0 \in [a, b]$. Then for all $x \in [a, b]$,

$$f(x) = P_{n-1}(x) + \int_{x_0}^{x} f^{(n)}(t)\frac{(x-t)^{n-1}}{(n-1)!}\, dt,$$

where P_{n-1} is the $(n-1)$-th Taylor polynomial of f.

Theorem 6.5.2 (Taylor Expansion with Big-$\mathcal{O}$ Remainder Term)
Let $n \geq 1$, $f : [a, b] \to \mathbb{R}$ an n-times continuously differentiable function and let $x_0 \in [a, b]$. Then for all $x \in [a, b]$

$$f(x) = P_{n-1} + \mathcal{O}(|x - x_0|^n) \qquad \text{as } x \to x_0.$$

Theorem 6.5.3 (Taylor Expansion with Little-o Remainder Term)
Let $n \geq 1$, $f : [a, b] \to \mathbb{R}$ an n-times continuously differentiable function and let $x_0 \in [a, b]$. Then for all $x \in [a, b]$

$$f(x) = P_n + o(|x - x_0|^n) \qquad \text{as } x \to x_0.$$

Remark 6.5.4
The crucial difference between these two remainder term representations is that for the stronger little-o statement we need a Taylor polynomial of order n, whereas for Big-$\mathcal{O}$ only order $(n-1)$ is required.

Remark 6.5.5 (Finding Taylor Expansions)
The Taylor expansion is unique. So if we find a polynomial (or a power series) that describes our function arbitrarily well in a neighborhood (i.e., has a little-o remainder term), then it must be the Taylor polynomial.

Example 6.5.6 (Taylor in Practice)
We seek the Taylor polynomial of $e^{(x^2)}$ up to order 4. For this we can simply substitute x^2 into the Taylor expansion of e^x. We thus obtain

$$e^{(x^2)} = 1 + (x^2) + \frac{(x^2)^2}{2} + \mathcal{O}((x^2)^3) = 1 + x^2 + \frac{x^4}{2} + \mathcal{O}(x^6).$$

If on the other hand we seek the Taylor polynomial of $(e^x)^2$ up to order 3, we can multiply the Taylor polynomials of e^x together. Thus we obtain

3

$$(e^x)^2 = \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \mathcal{O}(x^4)\right)\left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \mathcal{O}(x^4)\right)$$
$$= 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + x + x^2 + \frac{x^3}{2} + \frac{x^2}{2} + \frac{x^3}{2} + \frac{x^3}{6} + \mathcal{O}(x^4) = 1 + 2x + 2x^2 + \frac{4}{3}x^3 + \mathcal{O}(x^4).$$

Example 6.5.7 (Taylor Expansion of $\log(1-x)$)
We seek the Taylor expansion of $\log(1 - x)$ without computing all derivatives. For this we use that the Taylor expansion is unique. We know that

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n.$$

Now integrating both sides (the power series termwise by definition), we obtain

$$-\log(1-x) = \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1} = \sum_{n=1}^{\infty} \frac{x^n}{n}.$$

Thus for the Taylor expansion of $\log(1 - x)$ we have

$$\log(1-x) = -\sum_{n=1}^{\infty} \frac{x^n}{n} = -x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots$$

Example 6.5.8 (Limits using Taylor Expansion)
We can often compute limits towards a point simply by substituting the Taylor expansion (here it is important that the expansion point corresponds to the limit point). For example with this limit:

$$\lim_{x\to 0}\frac{\sin x - x}{x(1-\cos x)} = \lim_{x\to 0}\frac{-\frac{1}{6}x^3+\mathcal{O}(x^5)}{x(\frac{x^2}{2}+\mathcal{O}(x^4))} = \lim_{x\to 0}\frac{-\frac{1}{6}+\mathcal{O}(x^2)}{\frac{1}{2}+\mathcal{O}(x^2)} = \frac{-\frac{1}{6}}{\frac{1}{2}} = -\frac{1}{3}.$$

Example 6.5.9 (FS2024, A. Figalli)
Compute the limit.

$$\lim_{x\to 0}\frac{x+\cos(x)\sin(3x)}{\sin^3(x)+3\sin(x)-x}.$$

Solution: We substitute the Taylor expansions and find

$$\begin{aligned}\lim_{x\to 0}\frac{x+\cos(x)\sin(3x)}{\sin^3(x)+3\sin(x)-x} &= \lim_{x\to 0}\frac{x+\big(1+o(x^2)\big)\big(3x+o(x^2)\big)}{x^3+o(x^3)+3x+o(x^2)-x}\\ &= \lim_{x\to 0}\frac{x+3x+o(x^2)}{2x+o(x^2)} = \lim_{x\to 0}\frac{4x}{2x}+\lim_{x\to 0}o(x^2) = 2.\end{aligned}$$

Example 6.5.10 („Mean Girls" – Limit)
We seek the following limit

$$\lim_{x\to 0}\frac{\log(1-x)-\sin(x)}{1-\cos^2(x)}.$$

3

Solution: Using Taylor expansion, we can now investigate the behavior of the term near the singularity:

$$\lim_{x\to 0}\frac{\log(1-x)-\sin(x)}{1-\cos^2(x)} = \lim_{x\to 0}\frac{-\left(x+\frac{x^2}{2}+\frac{x^3}{3}+o(x^3)\right)-\left(x+\frac{x^3}{6}+o(x^3)\right)}{1-\left(1-\frac{x^2}{2}+o(x^3)\right)^2}$$

$$\lim_{x\to 0}\frac{-2x-\frac{x^2}{2}-\frac{x^3}{2}+o(x^3)}{x^2+o(x^3)} = \lim_{x\to 0}\frac{-2-\frac{x}{2}-\frac{x^2}{2}+o(x^3)}{x+o(x^3)} \sim \lim_{x\to 0}\left(\frac{-2}{x}-\frac{1}{2}\right)$$

Thus near zero, the term behaves like $\frac{-2}{x}-\frac{1}{2}$. This shows that the left-hand and right-hand limits are different and thus the limit does not exist.

Theorem 6.5.11 (Taylor with Lagrange Remainder Term)
Let $n \geq 1$, $f : [a,b] \to \mathbb{R}$ an n-times continuously differentiable function and let $x_0 \in [a,b]$. Then for all $x \in [a,b]$ there exists a $\xi_L \in (x_0, x)$ such that the Lagrange remainder term formula holds:

$$f(x) - P_{n-1}(x) = \frac{1}{n!}f^{(n)}(\xi_L)(x-x_0)^n.$$

6.5.2 Analytic Functions

Definition 6.5.12 (Analytic Function)
Let $I \subseteq \mathbb{R}$ be an interval and $x_0 \in I$. A smooth function $f : I \to \mathbb{R}$ is called *analytic at the point* x_0 if there exists $\delta > 0$ such that the Taylor series of f at x_0 has a radius of convergence $R > \delta$ and

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n \qquad \forall x \in (x_0 - \delta, x_0 + \delta) \cap I.$$

We say that f is *analytic* on I if f is analytic at all points in I.

Remark 6.5.13
In other words, an analytic function $f : I \to \mathbb{R}$ is characterized by the fact that for all points $x_0 \in I$ there exists a power series that converges pointwise in a neighborhood of x_0 to f.

Remark 6.5.14
Being infinitely differentiable and being analytic are not equivalent. There exist smooth (C^∞) functions that are generally not analytic! There even exist such smooth functions that are **nowhere** analytic.

Theorem 6.5.15 (A Criterion for Analyticity (Exam))
Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ a smooth function. Given $x_0 \in I$, suppose there exist constants $r, c, A > 0$ such that

$$|f^{(n)}(x_0)| < cA^n n! \qquad \text{for all } x \in (x_0 - r, x_0 + r) \cap I, n \in \mathbb{N}$$

Then f is analytic at x_0.

3

Proof. We consider the Taylor series: $\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$ and show that it equals $f(x)$. Now we know that the n-th derivative is bounded and can thus conclude for the radius of convergence of the power series:

$$\limsup_{n\to\infty} \sqrt[n]{\frac{f^{(n)}(x_0)}{n!}} \leq \limsup_{n\to\infty} \sqrt[n]{\frac{cA^n n!}{n!}} \leq \limsup_{n\to\infty} A \sqrt[n]{c} = A$$

Thus by definition, the radius of convergence $R \geq \frac{1}{A}$.

Now we choose $\delta < \min\{r, \frac{1}{A}\}$, so that we always remain within the interval I. Now consider the Lagrange remainder term of the Taylor polynomial of order $(n-1)$.

$$|f(x) - P_{n-1}(x)| = \frac{1}{n!}|f^{(n)}(\xi_L)||x - x_0|^n \leq cA^n \underbrace{|x - x_0|^n}_{<\delta} < c(A\delta)^n \xrightarrow{n\to\infty} 0$$

We know that δ can be at most $\frac{1}{A}$ by our construction. Therefore $A\delta < 1$, which means that as $n \to \infty$

$$f(x) = \lim_{n\to\infty} P_{n-1} = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n,$$

which was to be proved. □

Chapter 7

Ordinary Differential Equations

7.1 Ordinary Differential Equation

Definition 7.1.1 (Ordinary Differential Equation)
An ordinary differential equation is a relation between a function $u : \mathbb{R} \to \mathbb{R}$ and its derivatives. The general form of an n-th order differential equation is

$$G\left(x, u(x), u'(x), u''(x), \ldots, u^{(n)}(x)\right) = 0,$$

where $G : \mathbb{R}^{n+1} \to \mathbb{R}$ is a given function.
A **homogeneous** differential equation has no dependence on x, but only on u and its derivatives.

7.2 Solution Methods

7.2.1 Separation of Variables

Separation of variables can be used to solve so-called **separable** first-order differential equations.
4 A first-order differential equation is separable if it can be brought into the form

$$y'(x) = f(y)g(x).$$

Example 7.2.1 (Separable ODE)
An example of a separable differential equation is

$$y' = xy^2 + x = x(y^2 + 1).$$

Example 7.2.2 (Non-separable ODE)
The differential equation $y' = x^2 y + 1$ is *not* separable.

Remark 7.2.3 (Derivation of Separation of Variables)
Consider a separable differential equation of the form

$$y' = f(y)g(x).$$

Now define $h(t) = \frac{1}{f(t)}$ and write

$$\int h(y(x))y'(x)dx = \int g(x)dx.$$

By the substitution rule (1st form) we have

$$\int \frac{1}{f(y)}dy = \int g(x)dx.$$

Now let H be an antiderivative of $\frac{1}{f}$ and G an antiderivative of g, then

$$H(y(x)) = G(x) + C \iff y(x) = H^{-1}(G(x) + C).$$

The boundary condition of the problem determines the constant C.

Example 7.2.4 (Separation of Variables)
Let the differential equation $y' = xy^2 + x$ from the above example be given. First we bring the differential equation into a separated form

$$\frac{dy}{dx} = x(y^2 + 1).$$

In practice, the *Leibniz notation* is very useful. We separate the variables

$$\frac{1}{y^2+1}\, dy = x\, dx,$$

4 and then integrate both sides

$$\int \frac{1}{y^2+1}\, dy = \int x\, dx.$$

Our rigorous proof above justifies this transformation.
Thus the solution is

$$\arctan(y) = \frac{1}{2}x^2 + C \iff y(x) = \tan\left(\frac{1}{2}x^2 + C\right).$$

The boundary condition determines the constant C.

7.2.2 Exponential Ansatz

First we only consider **homogeneous** differential equations, that means there are no terms that do not depend on our solution function. **Inhomogeneous** problems will be considered in the next section.

With this ansatz we consider linear homogeneous differential equations of order n with constant coefficients.

$$a_n y^{(n)}(x) + \cdots + a_1 y'(x) + a_0 y(x) = 0$$

We now choose an ansatz $Ae^{\lambda x}$. Substituting this ansatz into the differential equation, we obtain its *characteristic polynomial*. The zeros of this polynomial are the possible values for the λ_i in our linear combination. The constants are determined by the boundary conditions.

Real Roots

(1)

$$y'(x) = 3y(x) \rightsquigarrow \lambda = 3.$$

Thus we know our solution as

$$y(x) = Ae^{3x}.$$

(2)

$$y^{(3)}(x) - 5y''(x) + 8y'(x) - 4y(x) = 0 \rightsquigarrow \lambda^3 - 5\lambda^2 + 8\lambda - 4 = (\lambda - 1)(\lambda - 2)^2 = 0$$

The roots of the characteristic polynomial are 1 and 2 as a double root. Therefore we must include an additional power of x for $\lambda = 2$. The solution thus looks like

$$y(x) = Ae^x + Be^{2x} + Cxe^{2x}.$$

Remark 7.2.5 (Note on example (2):)
For roots of higher order we include additional powers of x. For a root of third order one would additionally include a term with x^2 and so on.

Example: Exponential Ansatz (Real Roots)

Let the differential equation $y'' + 6y' + 9y = 0$ be given. We determine the roots of the characteristic polynomial

$$\lambda^2 + 6\lambda + 9 = 0 \rightsquigarrow (x + 3)^2 = 0.$$

4

Since $\lambda = -3$ is a double root, the solution ansatz is

$$y(x) = Ae^{-3x} + Bxe^{-3x}$$

Complex Roots

Complex roots always occur in pairs with their complex conjugate. If we have such a complex pair, we must extend our ansatz in the solution with sine and cosine terms. If we have a pair of complex roots $\lambda = \alpha \pm i\beta$, then we add in our solution ansatz

$$Ae^{\alpha x}\cos(\beta x) + Be^{\alpha x}\sin(\beta x)$$

If in this case we have complex roots of higher order, we again simply include additional powers of x.

Example: Exponential Ansatz (Complex Roots)

We consider the harmonic oscillator. Its differential equation is given by

$$y'' + \omega^2 y = 0.$$

The characteristic polynomial is thus

$$\lambda^2 = -\omega^2$$

with the unique solutions

$$\lambda = \pm i\omega \rightsquigarrow \alpha = 0 \text{ and } \beta = \omega.$$

Therefore the solution is

$$y(x) = Ae^{\alpha x}\cos(\beta x) + Be^{\alpha x}\sin(\beta x) = A\cos(\omega x) + B\sin(\omega x).$$

The constants A and B are determined by the boundary conditions.

7.2.3 Inhomogeneous Differential Equations and Particular Solutions

With this ansatz we consider linear *inhomogeneous* differential equations of order n with constant coefficients

$$a_n y^{(n)}(x) + \cdots + a_1 y'(x) + a_0 y(x) = g(x),$$

where now the so-called *forcing function* $g(x) \neq 0$.

Here we split the solution function $y(x)$ into two parts, a homogeneous solution $y_{\text{Hom.}}$ and a particular solution $y_{\text{Part.}}$

$$y(x) = y_{\text{Hom.}} + y_{\text{Part.}}.$$

Since we know that $y_{\text{Hom.}}$ solves the homogeneous equation, we have

$$a_n y^{(n)}_{\text{Hom.}}(x) + \cdots + a_1 y'_{\text{Hom.}}(x) + a_0 y_{\text{Hom.}}(x) = 0,$$

so this part simply vanishes and only the particular part remains with

$$a_n y^{(n)}_{\text{Part.}}(x) + \cdots + a_1 y'_{\text{Part.}}(x) + a_0 y_{\text{Part.}}(x) = g(x).$$

4

Because the differential equation is linear, our linear combination of both partial solutions $y(x) = y_{\text{Hom.}} + y_{\text{Part.}}$ is a complete solution.

Example: Inhomogeneous Differential Equation with Polynomial Forcing Term

Let the differential equation $y''(x) + 3y(x) + x^2 = 0$ be given. We see that the $+x^2$ term makes the problem inhomogeneous. First we solve the homogeneous problem

$$y''_{\text{Hom.}}(x) + 3y_{\text{Hom.}}(x) = 0 \rightsquigarrow \lambda^2 + 3 = 0$$

Thus the roots are $\lambda = \pm\sqrt{3}i$. So we have found our homogeneous solution as

$$y_{\text{Hom.}}(x) = A\sin(\sqrt{3}x) + B\cos(\sqrt{3}x).$$

Now we choose a polynomial ansatz as the particular solution. We set

$$y_{\text{Part.}} = Cx^2 + Dx + E$$

into the differential equation and obtain

$$2C + 3Cx^2 + 3Dx + 3E + x^2 = 0.$$

From this, by comparing coefficients, we obtain three equations

$$3C + 1 = 0 \qquad 3D = 0 \qquad 2C + 3E = 0.$$

Solving this system yields the particular solution as

$$y_{\text{Part.}} = -\frac{1}{3}x^2 + \frac{2}{9}.$$

Thus the complete solution of the inhomogeneous differential equation is

$$y(x) = A\sin(\sqrt{3}x) + B\cos(\sqrt{3}x) - \frac{1}{3}x^2 + \frac{2}{9}.$$

Example: Inhomogeneous Differential Equation with Trigonometric Forcing Term

Let the differential equation $y''(x) + 4y(x) = \sin(2x)$ be given. The $\sin(2x)$ term makes the problem inhomogeneous. First we solve the homogeneous problem

$$y''_{\text{Hom.}}(x) + 4y_{\text{Hom.}}(x) = 0 \rightsquigarrow \lambda^2 + 4 = 0.$$

Thus the roots are $\lambda = \pm 2i$. Therefore our homogeneous solution is

$$y_{\text{Hom.}}(x) = A\sin(2x) + B\cos(2x).$$

Now we choose an ansatz for the particular solution. Since the forcing term $\sin(2x)$ is trigonometric, we set

$$y_{\text{Part.}} = Cx\sin(2x) + Dx\cos(2x).$$

We substitute $y_{\text{Part.}}$ into the differential equation and by comparing coefficients we obtain

$$y_{\text{Part.}} = -\frac{1}{4}\cos(2x).$$

The complete solution of the inhomogeneous differential equation is therefore

4

$$y(x) = A\sin(2x) + B\cos(2x) - \frac{1}{4}\cos(2x).$$

Example: Inhomogeneous Differential Equation with Exponential Forcing Term

Let the differential equation $y''(x) - 2y'(x) + y(x) = e^x$ be given. The e^x term makes the problem inhomogeneous. First we solve the homogeneous problem

$$y''_{\text{Hom.}}(x) - 2y'_{\text{Hom.}}(x) + y_{\text{Hom.}}(x) = 0 \rightsquigarrow \lambda^2 - 2\lambda + 1 = 0.$$

Thus the roots are $\lambda = 1$ (double root). Therefore our homogeneous solution is

$$y_{\text{Hom.}}(x) = (A + Bx)e^x.$$

Now we choose an ansatz for the particular solution. Since the forcing terms e^x and xe^x would be a solution of the homogeneous problem, we multiply the ansatz by x^2 to avoid linear dependence. So we choose as a particular ansatz

$$y_{\text{Part.}} = Cx^2e^x.$$

We compute the derivatives:

$$y'_{\text{Part.}} = (2Cx + Cx^2)e^x, \quad y''_{\text{Part.}} = (2C + 4Cx + Cx^2)e^x.$$

Substituting into the differential equation gives

$$(2C + 4Cx + Cx^2)e^x - 2((2Cx + Cx^2)e^x) + (Cx^2e^x) = e^x.$$

After simplification and comparing coefficients we obtain $C = \frac{1}{2}$. The particular solution is thus

$$y_{\text{Part.}} = \frac{1}{2}x^2 e^x.$$

The complete solution of the inhomogeneous differential equation is therefore

$$y(x) = (A + Bx)e^x + \frac{1}{2}x^2 e^x.$$

7.2.4 Variation of Constants

Variation of Constants is another method to find the particular solution for an inhomogeneous ODE with constant coefficients. The general idea is to find the homogeneous solution and then consider the constant coefficients due to integration as functions themselves. By plugging it into the ODE one finds an ODE in the parameter function and solving this results in the general solution of the inhomogeneous ODE.

Example 7.2.6
Consider the inhomogeneous first-order ODE

$$y'(t) - y(t) = e^{2t}.$$

First, we solve the homogeneous equation

$$y'(t) - y(t) = 0,$$

which has the general solution

$$y_h(t) = Ce^t,$$

4 which can both be found via Seperation of Variables as well as Euler's method. In the method of variation of constants, we replace the constant C by a function $C(t)$ and make the ansatz

$$y_p(t) = C(t)e^t.$$

Note now, that we replaces C with $C(t)$! Differentiating with respect to t yields

$$y_p'(t) = C'(t)e^t + C(t)e^t.$$

Substituting y_p and y_p' into the original equation gives

$$C'(t)e^t = e^{2t},$$

and hence

$$C'(t) = e^t.$$

This is our differential equation in C which we know have to solve. Integrating, we obtain $C(t) = e^t + D.$ with a constant of integration D.
Therefore, a particular solution is

$$y_p(t) = (e^t + D) \cdot e^t = e^{2t} + De^t.$$

The general solution of the inhomogeneous equation is thus

$$y(t) = De^t + e^{2t}.$$

We can clearly see the general solution structure again, where De^t is the homogenous solution and e^{2t} is the particular solution.